PART I

Groups and Subgroups

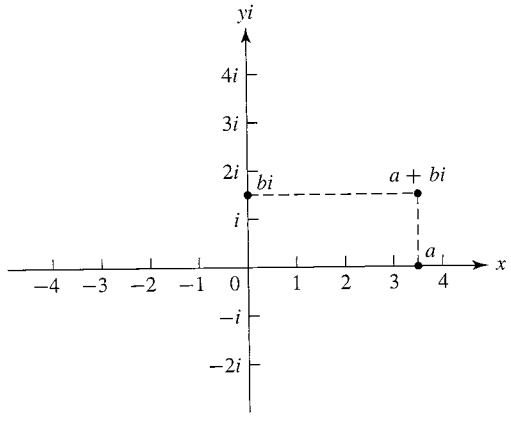
Section 1 Introduction and Examples
Section 2 Binary Operations
Section 3 Isomorphic Binary Structures
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SECTION 1 Introduction and Examples

In this section, we attempt to give you a little idea of the nature of abstract algebra. We are all familiar with addition and multiplication of real numbers. Both addition and multiplication combine two numbers to obtain one number. For example, addition combines 2 and 3 to obtain 5. We consider addition and multiplication to be *binary operations*. In this text, we abstract this notion, and examine sets in which we have one or more binary operations. We think of a binary operation on a set as giving an algebra on the set, and we are interested in the *structural properties* of that algebra. To illustrate what we mean by a structural property with our familiar set $\mathbb{R}$ of real numbers, note that the equation $x + x = a$ has a solution x in $\mathbb{R}$ for each $a \in \mathbb{R}$, namely, $x = a/2$. However, the corresponding multiplicative equation $x \cdot x = a$ does not have a solution in $\mathbb{R}$ if $a < 0$. Thus, $\mathbb{R}$ with addition has a different algebraic structure than $\mathbb{R}$ with multiplication.

Sometimes two different sets with what we naturally regard as very different binary operations turn out to have the same algebraic structure. For example, we will see in Section 3 that the set $\mathbb{R}$ with addition has the same algebraic structure as the set $\mathbb{R}^+$ of positive real numbers with multiplication!

This section is designed to get you thinking about such things informally. We will make everything precise in Sections 2 and 3. We now turn to some examples. Multiplication of complex numbers of magnitude 1 provides us with several examples that will be useful and illuminating in our work. We start with a review of complex numbers and their multiplication.



1.1 Figure

Complex Numbers

A real number can be visualized geometrically as a point on a line that we often regard as an x-axis. A complex number can be regarded as a point in the Euclidean plane, as shown in Fig. 1.1. Note that we label the vertical axis as the yi-axis rather than just the y-axis, and label the point one unit above the origin with i rather than 1. The point with Cartesian coordinates (a, b) is labeled $a + bi$ in Fig. 1.1. The set $\mathbb{C}$ of **complex numbers** is defined by

$$\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}\}.$$

We consider $\mathbb{R}$ to be a subset of the complex numbers by identifying a real number r with the complex number $r + 0i$. For example, we write $3 + 0i$ as 3 and $-\pi + 0i$ as $-\pi$ and $0 + 0i$ as 0. Similarly, we write $0 + 1i$ as i and $0 + si$ as si.

Complex numbers were developed after the development of real numbers. The complex number i was *invented* to provide a solution to the quadratic equation $x^2 = -1$, so we require that

$$i^2 = -1. \tag{1}$$

Unfortunately, i has been called an **imaginary number,** and this terminology has led generations of students to view the complex numbers with more skepticism than the real numbers. Actually, *all* numbers, such as $1, 3, \pi, -\sqrt{3}$, and i are inventions of our minds. There is no physical entity that *is* the number 1. If there were, it would surely be in a place of honor in some great scientific museum, and past it would file a steady stream of mathematicians, gazing at 1 in wonder and awe. A basic goal of this text is to show how we can invent solutions of polynomial equations when the coefficients of the polynomial may not even be real numbers!

Multiplication of Complex Numbers

The product $(a + bi)(c + di)$ is defined in the way it must be if we are to enjoy the familiar properties of real arithmetic and require that $i^2 = -1$, in accord with Eq. (1).

Namely, we see that we want to have

$$\begin{aligned}(a+bi)(c+di) &= ac + adi + bci + bdi^2 \\ &= ac + adi + bci + bd(-1) \\ &= (ac - bd) + (ad + bc)i.\end{aligned}$$

Consequently, we define multiplication of $z_1 = a + bc$ and $z_2 = c + di$ as

$$z_1 z_2 = (a + bi)(c + di) = (ac - bd) + (ad + bc)i, \tag{2}$$

which is of the form $r + si$ with $r = ac - bd$ and $s = ad + bc$. It is routine to check that the usual properties $z_1 z_2 = z_2 z_1$, $z_1(z_2 z_3) = (z_1 z_2)z_3$ and $z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$ all hold for all $z_1, z_2, z_3 \in \mathbb{C}$.

1.2 Example Compute $(2 - 5i)(8 + 3i)$.

Solution We don't memorize Eq. (2), but rather we compute the product as we did to motivate that equation. We have

$$(2 - 5i)(8 + 3i) = 16 + 6i - 40i + 15 = 31 - 34i. \quad \blacktriangle$$

To establish the geometric meaning of complex multiplication, we first define the **absolute value** $|a + bi|$ of $a + bi$ by

$$|a + bi| = \sqrt{a^2 + b^2}. \tag{3}$$

This absolute value is a nonnegative real number and is the distance from $a + bi$ to the origin in Fig. 1.1. We can now describe a complex number z in the polar-coordinate form

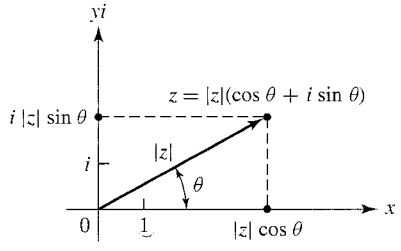
$$z = |z|(\cos\theta + i\sin\theta), \tag{4}$$

where θ is the angle measured counterclockwise from the x-axis to the vector from 0 to z, as shown in Fig. 1.3. A famous formula due to Leonard Euler states that

$$e^{i\theta} = \cos\theta + i\sin\theta.$$

Euler's Formula

We ask you to derive Euler's formula formally from the power series expansions for e^θ, $\cos\theta$ and $\sin\theta$ in Exercise 41. Using this formula, we can express z in Eq. (4) as



1.3 Figure

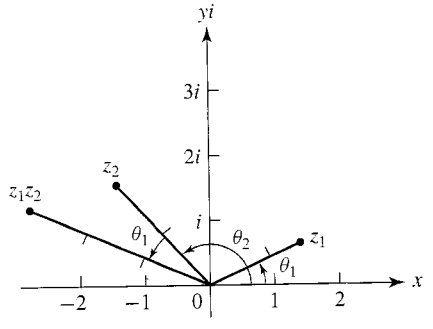
$z = |z|e^{i\theta}$. Let us set

$$z_1 = |z_1|e^{i\theta_1} \qquad \text{and} \qquad z_2 = |z_2|e^{i\theta_2}$$

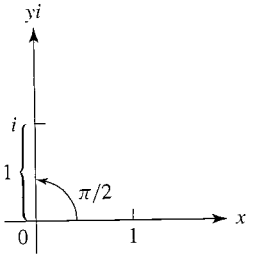
and compute their product in this form, assuming that the usual laws of exponentiation hold with complex number exponents. We obtain

$$\begin{aligned} z_1 z_2 &= |z_1|e^{i\theta_1}|z_2|e^{i\theta_2} = |z_1||z_2|e^{i(\theta_1+\theta_2)} \\ &= |z_1||z_2|[\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)]. \end{aligned} \tag{5}$$

Note that Eq. 5 concludes in the polar form of Eq. 4 where $|z_1 z_2| = |z_1||z_2|$ and the polar angle θ for $z_1 z_2$ is the sum $\theta = \theta_1 + \theta_2$. Thus, geometrically, we multiply complex numbers by multiplying their absolute values and adding their polar angles, as shown in Fig. 1.4. Exercise 39 indicates how this can be derived via trigonometric identities without recourse to Euler's formula and assumptions about complex exponentiation.



1.4 Figure



1.5 Figure

Note that i has polar angle $\pi/2$ and absolute value 1, as shown in Fig. 1.5. Thus i^2 has polar angle $2(\pi/2) = \pi$ and $|1 \cdot 1| = 1$, so that $i^2 = -1$.

1.6 Example Find all solutions in $\mathbb{C}$ of the equation $z^2 = i$.

Solution Writing the equation $z^2 = i$ in polar form and using Eq. (5), we obtain

$$|z|^2(\cos 2\theta + i\sin 2\theta) = 1(0 + i).$$

Thus $|z|^2 = 1$, so $|z| = 1$. The angle θ for z must satisfy $\cos 2\theta = 0$ and $\sin 2\theta = 1$. Consequently, $2\theta = (\pi/2) + n(2\pi)$, so $\theta = (\pi/4) + n\pi$ for an integer n. The values of n yielding values θ where $0 \le \theta < 2\pi$ are 0 and 1, yielding $\theta = \pi/4$ or $\theta = 5\pi/4$. Our solutions are

$$z_1 = 1\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right) \qquad \text{and} \qquad z_2 = 1\left(\cos\frac{5\pi}{4} + i\sin\frac{5\pi}{4}\right)$$

or

$$z_1 = \frac{1}{\sqrt{2}}(1+i) \qquad \text{and} \qquad z_2 = \frac{-1}{\sqrt{2}}(1+i). \qquad \blacktriangle$$

1.7 Example Find all solutions of $z^4 = -16$.

Solution As in Example 1.6 we write the equation in polar form, obtaining

$$|z|^4(\cos 4\theta + i \sin 4\theta) = 16(-1 + 0i).$$

Consequently, $|z|^4 = 16$, so $|z| = 2$ while $\cos 4\theta = -1$ and $\sin 4\theta = 0$. We find that $4\theta = \pi + n(2\pi)$, so $\theta = (\pi/4) + n(\pi/2)$ for integers n. The different values of θ obtained where $0 \leq \theta < 2\pi$ are $\pi/4, 3\pi/4, 5\pi/4$, and $7\pi/4$. Thus one solution of $z^4 = -16$ is

$$2\left(\cos\frac{\pi}{4} + i \sin\frac{\pi}{4}\right) = 2\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i\right) = \sqrt{2}(1 + i).$$

In a similar way, we find three more solutions,

$$\sqrt{2}(-1 + i), \qquad \sqrt{2}(-1 - i), \qquad \text{and} \qquad \sqrt{2}(1 - i). \qquad \blacktriangle$$

The last two examples illustrate that we can find solutions of an equation $z^n = a + bi$ by writing the equation in polar form. There will always be n solutions, provided that $a + bi \neq 0$. Exercises 16 through 21 ask you to solve equations of this type.

We will not use addition or division of complex numbers, but we probably should mention that addition is given by

$$(a + bi) + (c + di) = (a + c) + (b + d)i. \tag{6}$$

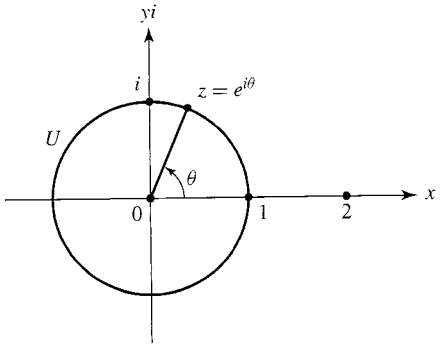
and division of $a + bi$ by nonzero $c + di$ can be performed using the device

$$\frac{a + bi}{c + di} = \frac{a + bi}{c + di} \cdot \frac{c - di}{c - di} = \frac{(ac + bd) + (bc - ad)i}{c^2 + d^2}$$
$$= \frac{ac + bd}{c^2 + d^2} + \frac{bc - ad}{c^2 + d^2}i. \tag{7}$$

Algebra on Circles

Let $U = \{z \in \mathbb{C} \mid |z| = 1\}$, so that U is the circle in the Euclidean plane with center at the origin and radius 1, as shown in Fig. 1.8. The relation $|z_1 z_2| = |z_1||z_2|$ shows that the product of two numbers in U is again a number in U; we say that U is *closed* under multiplication. Thus, we can view multiplication in U as providing algebra on the circle in Fig. 1.8.

As illustrated in Fig. 1.8, we associate with each $z = \cos\theta + i \sin\theta$ in U a real number $\theta \in \mathbb{R}$ that lies in the half-open interval where $0 \leq \theta < 2\pi$. This half-open interval is usually denoted by $[0, 2\pi)$, but we prefer to denote it by $\mathbb{R}_{2\pi}$ for reasons that will be apparent later. Recall that the angle associated with the product $z_1 z_2$ of two complex numbers is the sum $\theta_1 + \theta_2$ of the associated angles. Of course if $\theta_1 + \theta_2 \geq 2\pi$



1.8 Figure

then the angle in $\mathbb{R}_{2\pi}$ associated with $z_1 z_2$ is $\theta_1 + \theta_2 - 2\pi$. This gives us an **addition modulo 2π** on $\mathbb{R}_{2\pi}$. We denote this addition here by $+_{2\pi}$.

1.9 Example In $\mathbb{R}_{2\pi}$, we have $\frac{3\pi}{2} +_{2\pi} \frac{5\pi}{4} = \frac{11\pi}{4} - 2\pi = \frac{3\pi}{4}$. ▲

There was nothing special about the number 2π that enabled us to define addition on the half-open interval $\mathbb{R}_{2\pi}$. We can use any half-open interval $\mathbb{R}_c = \{x \in \mathbb{R} \mid 0 \leq x < c\}$.

1.10 Example In $\mathbb{R}_{23}$, we have $16 +_{23} 19 = 35 - 23 = 12$. In $\mathbb{R}_{8.5}$, we have $6 +_{8.5} 8 = 14 - 8.5 = 5.5$. ▲

Now complex number multiplication on the circle U where $|z| = 1$ and addition modulo 2π on $\mathbb{R}_{2\pi}$ have the same *algebraic properties*. We have the natural one-to-one correspondence $z \leftrightarrow \theta$ between $z \in U$ and $\theta \in \mathbb{R}_{2\pi}$ indicated in Fig. 1.8. Moreover, we deliberately defined $+_{2\pi}$ so that

$$\text{if} \quad z_1 \leftrightarrow \theta_1 \quad \text{and} \quad z_2 \leftrightarrow \theta_2, \quad \text{then} \quad z_1 \cdot z_2 \leftrightarrow (\theta_1 +_{2\pi} \theta_2). \tag{8}$$

isomorphism

The relation (8) shows that if we rename each $z \in U$ by its corresponding angle θ shown in Fig. 1.8, then the product of two elements in U is renamed by the sum of the angles for those two elements. Thus U with complex number multiplication and $\mathbb{R}_{2\pi}$ with addition modulo 2π must have the same algebraic properties. They differ only in the names of the elements and the names of the operations. Such a one-to-one correspondence satisfying the relation (8) is called an *isomorphism*. Names of elements and names of binary operations are not important in abstract algebra; we are interested in algebraic

properties. We illustrate what we mean by saying that the algebraic properties of U and of $\mathbb{R}_{2\pi}$ are the same.

1.11 Example In U there is exactly one element e such that $e \cdot z = z$ for all $z \in U$, namely, $e = 1$. The element 0 in $\mathbb{R}_{2\pi}$ that corresponds to $1 \in U$ is the only element e in $\mathbb{R}_{2\pi}$ such that $e +_{2\pi} x = x$ for all $x \in \mathbb{R}_{2\pi}$. ▲

1.12 Example The equation $z \cdot z \cdot z \cdot z = 1$ in U has exactly four solutions, namely, $1, i, -1$, and $-i$. Now $1 \in U$ and $0 \in \mathbb{R}_{2\pi}$ correspond, and the equation $x +_{2\pi} x +_{2\pi} x +_{2\pi} x = 0$ in $\mathbb{R}_{2\pi}$ has exactly four solutions, namely, $0, \pi/2, \pi$, and $3\pi/2$, which, of course, correspond to $1, i, -1$, and $-i$, respectively. ▲

Because our circle U has radius 1, it has circumference 2π and the radian measure of an angle θ is equal to the length of the arc the angle subtends. If we pick up our half-open interval $\mathbb{R}_{2\pi}$, put the 0 in the interval down on the 1 on the x-axis and wind it around the circle U counterclockwise, it will reach all the way back to 1. Moreover, each number in the interval will fall on the point of the circle having that number as the value of the central angle θ shown in Fig. 1.8. This shows that we could also think of addition on $\mathbb{R}_{2\pi}$ as being computed by adding lengths of subtended arcs counterclockwise, starting at $z = 1$, and subtracting 2π if the sum of the lengths is 2π or greater.

If we think of addition on a circle in terms of adding lengths of arcs from a starting point P on the circle and proceeding counterclockwise, we can use a circle of radius 2, which has circumference 4π, just as well as a circle of radius 1. We can take our half-open interval $\mathbb{R}_{4\pi}$ and wrap it around counterclockwise, starting at P; it will just cover the whole circle. Addition of arcs lengths gives us a notion of algebra for points on this circle of radius 2, which is surely isomorphic to $\mathbb{R}_{4\pi}$ with addition $+_{4\pi}$. However, if we take as the circle $|z| = 2$ in Fig. 1.8, multiplication of complex numbers does not give us an algebra on this circle. The relation $|z_1 z_2| = |z_1||z_2|$ shows that the product of two such complex numbers has absolute value 4 rather than 2. Thus complex number multiplication is *not closed* on this circle.

The preceding paragraphs indicate that a little geometry can sometimes be of help in abstract algebra. We can use geometry to convince ourselves that $\mathbb{R}_{2\pi}$ and $\mathbb{R}_{4\pi}$ are isomorphic. Simply stretch out the interval $\mathbb{R}_{2\pi}$ uniformly to cover the interval $\mathbb{R}_{4\pi}$, or, if you prefer, use a magnifier of power 2. Thus we set up the one-to-one correspondence $a \leftrightarrow 2a$ between $a \in \mathbb{R}_{2\pi}$ and $2a \in \mathbb{R}_{4\pi}$. The relation (8) for isomorphism becomes

$$\text{if} \quad a \leftrightarrow 2a \quad \text{and} \quad b \leftrightarrow 2b \quad \text{then} \quad (a +_{2\pi} b) \leftrightarrow (2a +_{4\pi} 2b). \tag{9}$$

isomorphism

This is obvious if $a + b \leq 2\pi$. If $a + b = 2\pi + c$, then $2a + 2b = 4\pi + 2c$, and the final pairing in the displayed relation becomes $c \leftrightarrow 2c$, which is true.

1.13 Example $x +_{4\pi} x +_{4\pi} x +_{4\pi} x = 0$ in $\mathbb{R}_{4\pi}$ has exactly four solutions, namely, $0, \pi, 2\pi$, and 3π, which are two times the solutions found for the analogous equation in $\mathbb{R}_{2\pi}$ in Example 1.12. ▲

There is nothing special about the numbers 2π and 4π in the previous argument. Surely, $\mathbb{R}_c$ with $+_c$ is isomorphic to $\mathbb{R}_d$ with $+_d$ for all $c, d \in \mathbb{R}^+$. We need only pair $x \in \mathbb{R}_c$ with $(d/c)x \in \mathbb{R}_d$.

Roots of Unity

The elements of the set $U_n = \{z \in \mathbb{C} \mid z^n = 1\}$ are called the **n^{th} roots of unity.** Using the technique of Examples 1.6 and 1.7, we see that the elements of this set are the numbers

$$\cos\left(m\frac{2\pi}{n}\right) + i \sin\left(m\frac{2\pi}{n}\right) \qquad \text{for} \qquad m = 0, 1, 2, \cdots, n-1.$$

They all have absolute value 1, so $U_n \subset U$. If we let $\zeta = \cos\frac{2\pi}{n} + i \sin\frac{2\pi}{n}$, then these n^{th} roots of unity can be written as

$$1 = \zeta^0, \zeta^1, \zeta^2, \zeta^3, \cdots, \zeta^{n-1}. \tag{10}$$

Because $\zeta^n = 1$, these n powers of ζ are closed under multiplication. For example, with $n = 10$, we have

$$\zeta^6\zeta^8 = \zeta^{14} = \zeta^{10}\zeta^4 = 1 \cdot \zeta^4 = \zeta^4.$$

Thus we see that we can compute $\zeta^i\zeta^j$ by computing $i +_n j$, viewing i and j as elements of $\mathbb{R}_n$.

Let $\mathbb{Z}_n = \{0, 1, 2, 3, \cdots, n-1\}$. We see that $\mathbb{Z}_n \subset \mathbb{R}_n$ and clearly addition modulo n is closed on $\mathbb{Z}_n$.

1.14 Example The solution of the equation $x + 5 = 3$ in $\mathbb{Z}_8$ is $x = 6$, because $5 +_8 6 = 11 - 8 = 3$. ▲

If we rename each of the nth roots of unity in (10) by its exponent, we use for names all the elements of $\mathbb{Z}_n$. This gives a one-to-one correspondence between U_n and $\mathbb{Z}_n$. Clearly,

$$\text{if} \quad \zeta^i \leftrightarrow i \quad \text{and} \quad \zeta^j \leftrightarrow j, \quad \text{then} \quad (\zeta^i \cdot \zeta^j) \leftrightarrow (i +_n j). \tag{11}$$

isomorphism

Thus U_n with complex number multiplication and $\mathbb{Z}_n$ with addition $+_n$ have the same algebraic properties.

1.15 Example It can be shown that there is an isomorphism of U_8 with $\mathbb{Z}_8$ in which $\zeta = e^{i2\pi/8} \leftrightarrow 5$. Under this isomorphism, we must then have $\zeta^2 = \zeta \cdot \zeta \leftrightarrow 5 +_8 5 = 2$. ▲

Exercise 35 asks you to continue the computation in Example 1.15, finding the elements of $\mathbb{Z}_8$ to which each of the remaining six elements of U_8 correspond.

EXERCISES 1

In Exercises 1 through 9 compute the given arithmetic expression and give the answer in the form $a + bi$ for $a, b \in \mathbb{R}$.

1. i^3 **2.** i^4 **3.** i^{23}

4. $(-i)^{35}$ **5.** $(4 - i)(5 + 3i)$ **6.** $(8 + 2i)(3 - i)$

7. $(2 - 3i)(4 + i) + (6 - 5i)$ **8.** $(1 + i)^3$ **9.** $(1 - i)^5$ (Use the binomial theorem.)

10. Find $|3 - 4i|$. **11.** Find $|6 + 4i|$.

In Exercises 12 through 15 write the given complex number z in the polar form $|z|(p + qi)$ where $|p + qi| = 1$.

12. $3 - 4i$ **13.** $-1 + i$ **14.** $12 + 5i$ **15.** $-3 + 5i$

In Exercises 16 through 21, find all solutions in $\mathbb{C}$ of the given equation.

16. $z^4 = 1$ **17.** $z^4 = -1$ **18.** $z^3 = -8$ **19.** $z^3 = -27i$

20. $z^6 = 1$ **21.** $z^6 = -64$

In Exercises 22 through 27, compute the given expression using the indicated modular addition.

22. $10 +_{17} 16$ **23.** $8 +_{10} 6$ **24.** $20.5 +_{25} 19.3$

25. $\frac{1}{2} +_1 \frac{7}{8}$ **26.** $\frac{3\pi}{4} +_{2\pi} \frac{3\pi}{2}$ **27.** $2\sqrt{2} +_{\sqrt{32}} 3\sqrt{2}$

28. Explain why the expression $5 +_6 8$ in $\mathbb{R}_6$ makes no sense.

In Exercises 29 through 34, find *all* solutions x of the given equation.

29. $x +_{15} 7 = 3$ in $\mathbb{Z}_{15}$ **30.** $x +_{2\pi} \frac{3\pi}{2} = \frac{3\pi}{4}$ in $\mathbb{R}_{2\pi}$

31. $x +_7 x = 3$ in $\mathbb{Z}_7$ **32.** $x +_7 x +_7 x = 5$ in $\mathbb{Z}_7$

33. $x +_{12} x = 2$ in $\mathbb{Z}_{12}$ **34.** $x +_4 x +_4 x +_4 x = 0$ in $\mathbb{Z}_4$

35. Example 1.15 asserts that there is an isomorphism of U_8 with $\mathbb{Z}_8$ in which $\zeta = e^{i(\pi/4)} \leftrightarrow 5$ and $\zeta^2 \leftrightarrow 2$. Find the element of $\mathbb{Z}_8$ that corresponds to each of the remaining six elements ζ^m in U_8 for $m = 0, 3, 4, 5, 6,$ and 7.

36. There is an isomorphism of U_7 with $\mathbb{Z}_7$ in which $\zeta = e^{i(2\pi/7)} \leftrightarrow 4$. Find the element in $\mathbb{Z}_7$ to which ζ^m must correspond for $m = 0, 2, 3, 4, 5,$ and 6.

37. Why can there be no isomorphism of U_6 with $\mathbb{Z}_6$ in which $\zeta = e^{i(\pi/3)}$ corresponds to 4?

38. Derive the formulas

$$\sin(a + b) = \sin a \cos b + \cos a \sin b$$

and

$$\cos(a + b) = \cos a \cos b - \sin a \sin b$$

by using Euler's formula and computing $e^{ia}e^{ib}$.

39. Let $z_1 = |z_1|(\cos\theta_1 + i \sin\theta_1)$ and $z_2 = |z_2|(\cos\theta_2 + i \sin\theta_2)$. Use the trigonometric identities in Exercise 38 to derive $z_1 z_2 = |z_1||z_2|[\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$.

40. **a.** Derive a formula for $\cos 3\theta$ in terms of $\sin\theta$ and $\cos\theta$ using Euler's formula.

b. Derive the formula $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ from part (a) and the identity $\sin^2\theta + \cos^2\theta = 1$. (We will have use for this identity in Section 32.)

41. Recall the power series expansions

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots + \frac{x^n}{n!} + \cdots,$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots + (-1)^{n-1}\frac{x^{2n-1}}{(2n-1)!} + \cdots, \text{ and}$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots + (-1)^n\frac{x^{2n}}{(2n)!} + \cdots$$

from calculus. Derive Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$ formally from these three series expansions.

SECTION 2 BINARY OPERATIONS

Suppose that we are visitors to a strange civilization in a strange world and are observing one of the creatures of this world drilling a class of fellow creatures in addition of numbers. Suppose also that we have not been told that the class is learning to add, but were just placed as observers in the room where this was going on. We are asked to give a report on exactly what happens. The teacher makes noises that sound to us approximately like *gloop, poyt.* The class responds with *bimt.* The teacher then gives *ompt, gaft,* and the class responds with *poyt.* What are they doing? We cannot report that they are adding numbers, for we do not even know that the sounds are representing numbers. Of course, we do realize that there is communication going on. All we can say with any certainty is that these creatures know some rule, so that when certain pairs of things are designated in their language, one after another, like *gloop, poyt,* they are able to agree on a response, *bimt.* This same procedure goes on in addition drill in our first grade classes where a teacher may say *four, seven,* and the class responds with *eleven.*

In our attempt to analyze addition and multiplication of numbers, we are thus led to the idea that addition is basically just a rule that people learn, enabling them to associate, with two numbers in a given order, some number as the answer. Multiplication is also such a rule, but a different rule. Note finally that in playing this game with students, teachers have to be a little careful of what two things they give to the class. If a first grade teacher suddenly inserts *ten, sky,* the class will be very confused. The rule is only defined for pairs of things from some specified set.

Definitions and Examples

As mathematicians, let us attempt to collect the core of these basic ideas in a useful definition, generalizing the notions of addition and multiplication of numbers. As we remarked in Section 0, we do not attempt to define a set. However, we can attempt to be somewhat mathematically precise, and we describe our generalizations as *functions* (see Definition 0.10 and Example 0.11) rather than as *rules.* Recall from Definition 0.4 that for any set S, the set $S \times S$ consists of all ordered pairs (a, b) for elements a and b of S.

2.1 Definition A **binary operation** $*$ on a set S is a function mapping $S \times S$ into S. For each $(a, b) \in S \times S$, we will denote the element $*((a, b))$ of S by $a * b$. ■

Intuitively, we may regard a binary operation $*$ on S as assigning, to each ordered pair (a, b) of elements of S, an element $a * b$ of S. We proceed with examples.

2.2 Example Our usual addition $+$ is a binary operation on the set $\mathbb{R}$. Our usual multiplication $\cdot$ is a different binary operation on $\mathbb{R}$. In this example, we could replace $\mathbb{R}$ by any of the sets $\mathbb{C}$, $\mathbb{Z}$, $\mathbb{R}^+$, or $\mathbb{Z}^+$. ▲

Note that we require a binary operation on a set S to be defined for *every* ordered pair (a, b) of elements from S.

2.3 Example Let $M(\mathbb{R})$ be the set of all matrices[†] with real entries. The usual matrix addition $+$ is *not* a binary operation on this set since $A + B$ is not defined for an ordered pair (A, B) of matrices having different numbers of rows or of columns. ▲

Sometimes a binary operation on S provides a binary operation on a subset H of S also. We make a formal definition.

2.4 Definition Let $*$ be a binary operation on S and let H be a subset of S. The subset H is **closed under** $*$ if for all $a, b \in H$ we also have $a * b \in H$. In this case, the binary operation on H given by restricting $*$ to H is the **induced operation** of $*$ on H. ■

By our very definition of a binary operation $*$ on S, the set S is closed under $*$, but a subset may not be, as the following example shows.

2.5 Example Our usual addition $+$ on the set $\mathbb{R}$ of real numbers does not induce a binary operation on the set $\mathbb{R}^*$ of nonzero real numbers because $2 \in \mathbb{R}^*$ and $-2 \in \mathbb{R}^*$, but $2 + (-2) = 0$ and $0 \notin \mathbb{R}^*$. Thus $\mathbb{R}^*$ is not closed under $*$. ▲

In our text, we will often have occasion to decide whether a subset H of S is closed under a binary operation $*$ on S. To arrive at a correct conclusion, *we have to know what it means for an element to be in* H, and to use this fact. Students have trouble here. Be sure you understand the next example.

2.6 Example Let $+$ and $\cdot$ be the usual binary operations of addition and multiplication on the set $\mathbb{Z}$, and let $H = \{n^2 | n \in \mathbb{Z}^+\}$. Determine whether H is closed under (a) addition and (b) multiplication.

For part (a), we need only observe that $1^2 = 1$ and $2^2 = 4$ are in H, but that $1 + 4 = 5$ and $5 \notin H$. Thus H is not closed under addition.

For part (b), suppose that $r \in H$ and $s \in H$. Using what it means for r and s to be in H, we see that there must be integers n and m in $\mathbb{Z}^+$ such that $r = n^2$ and $s = m^2$. Consequently, $rs = n^2m^2 = (nm)^2$. By the characterization of elements in H and the fact that $nm \in \mathbb{Z}^+$, this means that $rs \in H$, so H is closed under multiplication. ▲

[†] Most students of abstract algebra have studied linear algebra and are familiar with matrices and matrix operations. For the benefit of those students, examples involving matrices are often given. The reader who is not familiar with matrices can either skip all references to them or turn to the Appendix at the back of the text, where there is a short summary.

2.7 Example Let F be the set of all real-valued functions f having as domain the set $\mathbb{R}$ of real numbers. We are familiar from calculus with the binary operations $+$, $-$, $\cdot$, and $\circ$ on F. Namely, for each ordered pair (f, g) of functions in F, we define for each $x \in \mathbb{R}$

$$\begin{aligned} f+g \text{ by } (f+g)(x) &= f(x)+g(x) && \text{addition,} \\ f-g \text{ by } (f-g)(x) &= f(x)-g(x) && \text{subtraction,} \\ f\cdot g \text{ by } (f\cdot g)(x) &= f(x)g(x) && \text{multiplication,} \end{aligned}$$

and

$$f \circ g \text{ by } (f \circ g)(x) = f(g(x)) \qquad \text{composition.}$$

All four of these functions are again real valued with domain $\mathbb{R}$, so F is closed under all four operations $+$, $-$, $\cdot$, and $\circ$. ▲

The binary operations described in the examples above are very familiar to you. In this text, we want to *abstract* basic structural concepts from our familiar algebra. To emphasize this concept of *abstraction* from the familiar, we should illustrate these structural concepts with unfamiliar examples. We presented the binary operations of complex number multiplication on U and U_n, addition $+_n$ on $\mathbb{Z}_n$, and addition $+_c$ on $\mathbb{R}_c$ in Section 1.

The most important method of describing a particular binary operation $*$ on a given set is to characterize the element $a * b$ assigned to each pair (a, b) by some property defined in terms of a and b.

2.8 Example On $\mathbb{Z}^+$, we define a binary operation $*$ by $a * b$ equals the smaller of a and b, or the common value if $a = b$. Thus $2 * 11 = 2$; $15 * 10 = 10$; and $3 * 3 = 3$. ▲

2.9 Example On $\mathbb{Z}^+$, we define a binary operation $*'$ by $a *' b = a$. Thus $2 *' 3 = 2$, $25 *' 10 = 25$, and $5 *' 5 = 5$. ▲

2.10 Example On $\mathbb{Z}^+$, we define a binary operation $*''$ by $a *'' b = (a * b) + 2$, where $*$ is defined in Example 2.8. Thus $4 *'' 7 = 6$; $25 *'' 9 = 11$; and $6 *'' 6 = 8$. ▲

It may seem that these examples are of no importance, but consider for a moment. Suppose we go into a store to buy a large, delicious chocolate bar. Suppose we see two identical bars side by side, the wrapper of one stamped \$1.67 and the wrapper of the other stamped \$1.79. Of course we pick up the one stamped \$1.67. Our knowledge of which one we want depends on the fact that at some time we learned the binary operation $*$ of Example 2.8. It is a *very important operation*. Likewise, the binary operation $*'$ of Example 2.9 is defined using our ability to distinguish order. Think what a problem we would have if we tried to put on our shoes first, and then our socks! Thus we should not be hasty about dismissing some binary operation as being of little significance. Of course, our usual operations of addition and multiplication of numbers have a practical importance well known to us.

Examples 2.8 and 2.9 were chosen to demonstrate that a binary operation may or may not depend on the order of the given pair. Thus in Example 2.8, $a * b = b * a$ for all $a, b \in \mathbb{Z}^+$, and in Example 2.9 this is not the case, for $5 *' 7 = 5$ but $7 *' 5 = 7$.

2.11 Definition A binary operation $*$ on a set S is **commutative** if (and only if) $a * b = b * a$ for all $a, b \in S$. ■

As was pointed out in Section 0, it is customary in mathematics to omit the words *and only if* from a definition. Definitions are always understood to be if and only if statements. *Theorems are not always if and only if statements, and no such convention is ever used for theorems.*

Now suppose we wish to consider an expression of the form $a * b * c$. A binary operation $*$ enables us to combine only two elements, and here we have three. The obvious attempts to combine the three elements are to form either $(a * b) * c$ or $a * (b * c)$. With $*$ defined as in Example 2.8, $(2 * 5) * 9$ is computed by $2 * 5 = 2$ and then $2 * 9 = 2$. Likewise, $2 * (5 * 9)$ is computed by $5 * 9 = 5$ and then $2 * 5 = 2$. Hence $(2 * 5) * 9 = 2 * (5 * 9)$, and it is not hard to see that for this $*$,

$$(a * b) * c = a * (b * c),$$

so there is no ambiguity in writing $a * b * c$. But for $*''$ of Example 2.10,

$$(2 *'' 5) *'' 9 = 4 *'' 9 = 6,$$

while

$$2 *'' (5 *'' 9) = 2 *'' 7 = 4.$$

Thus $(a *'' b) *'' c$ need not equal $a *'' (b *'' c)$, and an expression $a *'' b *'' c$ may be ambiguous.

2.12 Definition A binary operation on a set S is **associative** if $(a * b) * c = a * (b * c)$ for all $a, b, c \in S$. ■

It can be shown that if $*$ is associative, then longer expressions such as $a * b * c * d$ are not ambiguous. Parentheses may be inserted in any fashion for purposes of computation; the final results of two such computations will be the same.

Composition of functions mapping $\mathbb{R}$ into $\mathbb{R}$ was reviewed in Example 2.7. For any set S and any functions f and g mapping S into S, we similarly define the composition $f \circ g$ of g followed by f as the function mapping S into S such that $(f \circ g)(x) = f(g(x))$ for all $x \in S$. Some of the most important binary operations we consider are defined using composition of functions. It is important to know that this composition is always associative whenever it is defined.

2.13 Theorem **(Associativity of Composition)** Let S be a set and let f, g, and h be functions mapping S into S. Then $f \circ (g \circ h) = (f \circ g) \circ h$.

Proof To show these two functions are equal, we must show that they give the same assignment to each $x \in S$. Computing we find that

$$(f \circ (g \circ h))(x) = f((g \circ h)(x)) = f(g(h(x)))$$

and

$$((f \circ g) \circ h)(x) = (f \circ g)(h(x)) = f(g(h(x))),$$

so the same element $f(g(h(x)))$ of S is indeed obtained. ◆

As an example of using Theorem 2.13 to save work, recall that it is a fairly painful exercise in summation notation to show that multiplication of $n \times n$ matrices is an associative binary operation. If, in a linear algebra course, we first show that there is a one-to-one correspondence between matrices and linear transformations and that multiplication of matrices corresponds to the composition of the linear transformations (functions), we obtain this associativity at once from Theorem 2.13.

Tables

For a finite set, a binary operation on the set can be defined by means of a table in which the elements of the set are listed across the top as heads of columns and at the left side as heads of rows. We always require that the elements of the set be listed as heads across the top in the same order as heads down the left side. The next example illustrates the use of a table to define a binary operation.

2.14 Example Table 2.15 defines the binary operation $*$ on $S = \{a, b, c\}$ by the following rule:

$$(i\text{th } \textit{entry on the left}) * (j\text{th } \textit{entry on the top})$$
$$= (\textit{entry in the } i\text{th } \textit{row and } j\text{th } \textit{column of the table body}).$$

2.15 Table

$*$	a	b	c
a	b	c	b
b	a	c	b
c	c	b	a

Thus $a * b = c$ and $b * a = a$, so $*$ is not commutative. ▲

We can easily see that *a binary operation defined by a table is commutative if and only if the entries in the table are symmetric with respect to the diagonal that starts at the upper left corner of the table and terminates at the lower right corner.*

2.16 Example Complete Table 2.17 so that $*$ is a commutative binary operation on the set $S = \{a, b, c, d\}$.

Solution From Table 2.17, we see that $b * a = d$. For $*$ to be commutative, we must have $a * b = d$ also. Thus we place d in the appropriate square defining $a * b$, which is located symmetrically across the diagonal in Table 2.18 from the square defining $b * a$. We obtain the rest of Table 2.18 in this fashion to give our solution. ▲

2.17 Table

$*$	a	b	c	d
a	b			
b	d	a		
c	a	c	d	
d	a	b	b	c

2.18 Table

$*$	a	b	c	d
a	b	d	a	a
b	d	a	c	b
c	a	c	d	b
d	a	b	b	c

Some Words of Warning

Classroom experience shows the chaos that may result if a student is given a set and asked to define some binary operation on it. Remember that in an attempt to define a binary operation $*$ on a set S we must be sure that

1. *exactly one element is assigned to each possible ordered pair of elements of* S,
2. *for each ordered pair of elements of* S, *the element assigned to it is again in* S.

Regarding Condition 1, a student will often make an attempt that assigns an element of S to "most" ordered pairs, but for a few pairs, determines no element. In this event, $*$ is **not everywhere defined** on S. It may also happen that for some pairs, the attempt could assign any of several elements of S, that is, there is ambiguity. In any case

of ambiguity, $*$ is **not well defined**. If Condition 2 is violated, then S is **not closed under** $*$.

Following are several illustrations of attempts to define binary operations on sets. Some of them are worthless. The symbol $*$ is used for the attempted operation in all these examples.

2.19 Example On $\mathbb{Q}$, let $a * b = a/b$. Here $*$ is *not everywhere defined* on $\mathbb{Q}$, for no rational number is assigned by this rule to the pair (2, 0). ▲

2.20 Example On $\mathbb{Q}^+$, let $a * b = a/b$. Here both Conditions 1 and 2 are satisfied, and $*$ is a binary operation on $\mathbb{Q}^+$. ▲

2.21 Example On $\mathbb{Z}^+$, let $a * b = a/b$. Here Condition 2 fails, for $1 * 3$ is not in $\mathbb{Z}^+$. Thus $*$ is not a binary operation on $\mathbb{Z}^+$, since $\mathbb{Z}^+$ is *not closed under* $*$. ▲

2.22 Example Let F be the set of all real-valued functions with domain $\mathbb{R}$ as in Example 2.7. Suppose we "define" $*$ to give the usual quotient of f by g, that is, $f * g = h$, where $h(x) = f(x)/g(x)$. Here Condition 2 is violated, for the functions in F were to be defined for *all* real numbers, and for some $g \in F$, $g(x)$ will be zero for some values of x in $\mathbb{R}$ and $h(x)$ would not be defined at those numbers in $\mathbb{R}$. For example, if $f(x) = \cos x$ and $g(x) = x^2$, then $h(0)$ is undefined, so $h \notin F$. ▲

2.23 Example Let F be as in Example 2.22 and let $f * g = h$, where h is the function greater than both f and g. This "definition" is completely worthless. In the first place, we have not defined what it means for one function to be greater than another. Even if we had, any sensible definition would result in there being many functions greater than both f and g, and $*$ would still be *not well defined.* ▲

2.24 Example Let S be a set consisting of 20 people, no two of whom are of the same height. Define $*$ by $a * b = c$, where c is the tallest person among the 20 in S. This is a perfectly good binary operation on the set, although not a particularly interesting one. ▲

2.25 Example Let S be as in Example 2.24 and let $a * b = c$, where c is the shortest person in S who is taller than both a and b. This $*$ is *not everywhere defined,* since if either a or b is the tallest person in the set, $a * b$ is not determined. ▲

EXERCISES 2

Computations

Exercises 1 through 4 concern the binary operation $*$ defined on $S = \{a, b, c, d, e\}$ by means of Table 2.26.

1. Compute $b * d$, $c * c$, and $[(a * c) * e] * a$.
2. Compute $(a * b) * c$ and $a * (b * c)$. Can you say on the basis of this computations whether $*$ is associative?
3. Compute $(b * d) * c$ and $b * (d * c)$. Can you say on the basis of this computation whether $*$ is associative?

2.26 Table

*	a	b	c	d	e
a	a	b	c	b	d
b	b	c	a	e	c
c	c	a	b	b	a
d	b	e	b	e	d
e	d	b	a	d	c

2.27 Table

*	a	b	c	d
a	a	b	c	
b	b	d		c
c	c	a	d	b
d	d			a

2.28 Table

*	a	b	c	d
a	a	b	c	d
b	b	a	c	d
c	c	d	c	d
d				

4. Is $*$ commutative? Why?

5. Complete Table 2.27 so as to define a commutative binary operation $*$ on $S = \{a, b, c, d\}$.

6. Table 2.28 can be completed to define an associative binary operation $*$ on $S = \{a, b, c, d\}$. Assume this is possible and compute the missing entries.

In Exercises 7 through 11, determine whether the binary operation $*$ defined is commutative and whether $*$ is associative.

7. $*$ defined on $\mathbb{Z}$ by letting $a * b = a - b$

8. $*$ defined on $\mathbb{Q}$ by letting $a * b = ab + 1$

9. $*$ defined on $\mathbb{Q}$ by letting $a * b = ab/2$

10. $*$ defined on $\mathbb{Z}^+$ by letting $a * b = 2^{ab}$

11. $*$ defined on $\mathbb{Z}^+$ by letting $a * b = a^b$

12. Let S be a set having exactly one element. How many different binary operations can be defined on S? Answer the question if S has exactly 2 elements; exactly 3 elements; exactly n elements.

13. How many different commutative binary operations can be defined on a set of 2 elements? on a set of 3 elements? on a set of n elements?

Concepts

In Exercises 14 through 16, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

14. A binary operation $*$ is *commutative* if and only if $a * b = b * a$.

15. A binary operation $*$ on a set S is *associative* if and only if, for all $a, b, c \in S$, we have $(b * c) * a = b * (c * a)$.

16. A subset H of a set S is *closed* under a binary operation $*$ on S if and only if $(a * b) \in H$ for all $a, b \in S$.

In Exercises 17 through 22, determine whether the definition of $*$ does give a binary operation on the set. In the event that $*$ is not a binary operation, state whether Condition 1, Condition 2, or both of these conditions on page 24 are violated.

17. On $\mathbb{Z}^+$, define $*$ by letting $a * b = a - b$.

18. On $\mathbb{Z}^+$, define $*$ by letting $a * b = a^b$.

19. On $\mathbb{R}$, define $*$ by letting $a * b = a - b$.

20. On $\mathbb{Z}^+$, define $*$ by letting $a * b = c$, where c is the smallest integer greater than both a and b.

21. On $\mathbb{Z}^+$, define $*$ by letting $a * b = c$, where c is at least 5 more than $a + b$.

22. On $\mathbb{Z}^+$, define $*$ by letting $a * b = c$, where c is the largest integer less than the product of a and b.

23. Let H be the subset of $M_2(\mathbb{R})$ consisting of all matrices of the form $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ for $a, b \in \mathbb{R}$. Is H closed under

a matrix addition? **b** matrix multiplication?

24. Mark each of the following true or false.

_____ **a.** If $*$ is any binary operation on any set S, then $a * a = a$ for all $a \in S$.

_____ **b.** If $*$ is any commutative binary operation on any set S, then $a * (b * c) = (b * c) * a$ for all $a, b, c \in S$.

_____ **c.** If $*$ is any associative binary operation on any set S, then $a * (b * c) = (b * c) * a$ for all $a, b, c \in S$.

_____ **d.** The only binary operations of any importance are those defined on sets of numbers.

_____ **e.** A binary operation $*$ on a set S is commutative if there exist $a, b \in S$ such that $a * b = b * a$.

_____ **f.** Every binary operation defined on a set having exactly one element is both commutative and associative.

_____ **g.** A binary operation on a set S assigns at least one element of S to each ordered pair of elements of S.

_____ **h.** A binary operation on a set S assigns at most one element of S to each ordered pair of elements of S.

_____ **i.** A binary operation on a set S assigns exactly one element of S to each ordered pair of elements of S.

_____ **j.** A binary operation on a set S may assign more than one element of S to some ordered pair of elements of S.

25. Give a set different from any of those described in the examples of the text and not a set of numbers. Define two different binary operations $*$ and $*'$ on this set. Be sure that your set is *well defined.*

Theory

26. Prove that if $*$ is an associative and commutative binary operation on a set S, then

$$(a * b) * (c * d) = [(d * c) * a] * b$$

for all $a, b, c, d \in S$. Assume the associative law only for triples as in the definition, that is, assume only

$$(x * y) * z = x * (y * z)$$

for all $x, y, z \in S$.

In Exercises 27 and 28, either prove the statement or give a counterexample.

27. Every binary operation on a set consisting of a single element in both commutative and associative.

28. Every commutative binary operation on a set having just two elements is associative.

Let F be the set of all real-valued functions having as domain the set $\mathbb{R}$ of all real numbers. Example 2.7 defined the binary operations $+$, $-$, $\cdot$, and $\circ$ on F. In Exercises 29 through 35, either prove the given statement or give a counterexample.

29. Function addition $+$ on F is associative.

30. Function subtraction $-$ on F is commutative

31. Function subtraction $-$ on F is associative.

32. Function multiplication $\cdot$ on F is commutative.

33. Function multiplication $\cdot$ on F is associative.

34. Function composition $\circ$ on F is commutative.

35. If $*$ and $*'$ are any two binary operations on a set S, then

$$a * (b *' c) = (a * b) *' (a * c) \qquad \text{for all} \quad a, b, c \in S.$$

36. Suppose that $*$ is an *associative binary* operation on a set S. Let $H = \{a \in S \mid a * x = x * a \text{ for all } x \in S\}$. Show that H is closed under $*$. (We think of H as consisting of all elements of S that *commute* with every element in S.)

37. Suppose that $*$ is an associative and commutative binary operation on a set S. Show that $H = \{a \in S \mid a * a = a\}$ is closed under $*$. (The elements of H are **idempotents** of the binary operation $*$.)

SECTION 3 Isomorphic Binary Structures

Compare Table 3.1 for the binary operation $*$ on the set $S = \{a, b, c\}$ with Table 3.2 for the binary operation $*'$ on the set $T = \{\#, \$, \&\}$.

Notice that if, in Table 3.1, we replace all occurrences of a by #, every b by \$, and every c by & using the one-to-one correspondence

$$a \leftrightarrow \# \qquad b \leftrightarrow \$ \qquad c \leftrightarrow \&$$

we obtain precisely Table 3.2. The two tables differ only in the symbols (or names) denoting the elements and the symbols $*$ and $*'$ for the operations. If we rewrite Table 3.3 with elements in the order y, x, z, we obtain Table 3.4. (Here we did not set up any one-one-correpondence; we just listed the same elements in different order outside the heavy bars of the table.) Replacing, in Table 3.1, all occurrences of a by y, every b by x, and every c by z using the one-to-one correspondence

$$a \leftrightarrow y \qquad b \leftrightarrow x \qquad c \leftrightarrow z$$

we obtain Table 3.4. We think of Tables 3.1, 3.2, 3.3, and 3.4 as being *structurally alike*. These four tables differ only in the names (or symbols) for their elements and in the order that those elements are listed as heads in the tables. However, Table 3.5 for binary operation $\bar{*}$ and Table 3.6 for binary operation $\hat{*}$ on the set $S = \{a, b, c\}$ are *structurally different* from each other and from Table 3.1. In Table 3.1, each element appears three times in the body of the table, while the body of Table 3.5 contains the single element b. In Table 3.6, for all $s \in S$ we get the same value c for $s \hat{*} s$ along the upper-left to lower-right diagonal, while we get three different values in Table 3.1. Thus Tables 3.1 through 3.6 give just three structurally different binary operations on a set of three elements, provided we disregard the names of the elements and the order in which they appear as heads in the tables.

The situation we have just discussed is somewhat akin to children in France and in Germany learning the operation of addition on the set $\mathbb{Z}^+$. The children have different

3.1 Table

$*$	a	b	c
a	c	a	b
b	a	b	c
c	b	c	a

3.2 Table

$*'$	#	\$	&
#	&	#	\$
\$	#	\$	&
&	\$	&	#

3.3 Table

$*''$	x	y	z
x	x	y	z
y	y	z	x
z	z	x	y

3.4 Table

$*''$	y	x	z
y	z	y	x
x	y	x	z
z	x	z	y

3.5 Table

$\bar{*}$	a	b	c
a	b	b	b
b	b	b	b
c	b	b	b

3.6 Table

$\hat{*}$	a	b	c
a	c	a	b
b	b	c	a
c	a	b	c

names (un, deux, trois, $\cdots$ versus ein, zwei, drei $\cdots$) for the numbers, but they are learning the same binary structure. (In this case, they are also using the same symbols for the numbers, so their addition tables would appear the same if they list the numbers in the same order.)

We are interested in studying the different types of *structures* that binary operations can provide on sets having the same number of elements, as typified by Tables 3.4, 3.5, and 3.6. Let us consider a **binary algebraic structure**[†] $\langle S, * \rangle$ to be a set S together with a binary operation $*$ on S. In order for two such binary structures $\langle S, * \rangle$ and $\langle S', *' \rangle$ to be structurally alike in the sense we have described, we would have to have a one-to-one correspondence between the elements x of S and the elements x' of S' such that

$$\text{if} \quad x \leftrightarrow x' \quad \text{and} \quad y \leftrightarrow y', \quad \text{then} \quad x * y \leftrightarrow x' *' y'. \tag{1}$$

A one-to-one correspondence exists if the sets S and S' have the same number of elements. It is customary to describe a one-to-one correspondence by giving a *one-to-one* function ϕ mapping S *onto* S' (see Definition 0.12). For such a function ϕ, we regard the equation $\phi(x) = x'$ as reading the one-to-one pairing $x \leftrightarrow x$ in left-to-right order. In terms of ϕ, the final $\leftrightarrow$ correspondence in (1), which asserts the algebraic structure in S' is the same as in S, can be expressed as

$$\phi(x * y) = \phi(x) *' \phi(y).$$

Such, a function showing that two algebraic systems are structurally alike is known as an *isomorphism*. We give a formal definition.

3.7 Definition Let $\langle S, * \rangle$ and $\langle S', *' \rangle$ be binary algebraic structures. An **isomorphism of** S with S' is a one-to-one function ϕ mapping S onto S' such that

$$\phi(x * y) = \phi(y) *' \phi(y) \text{ for all } x, y \in S. \tag{2}$$

homomorphism property

[†] Remember that boldface type indicates that a term is being defined.

If such a map ϕ exists, then S and S' are **isomorphic binary structures**, which we denote by $S \simeq S'$, omitting the $*$ and $*'$ from the notation. ■

You may wonder why we labeled the displayed condition in Definition 3.7 the *homomorphism property* rather than the *isomorphism property*. The notion of isomorphism includes the idea of one-to-one correspondence, which appeared in the definition via the words *one-to-one* and *onto* before the display. In Chapter 13, we will discuss the relation between S and S' when $\phi : S \to S'$ satisfies the displayed homomorphism property, but ϕ is not necessarily one to one; ϕ is then called a *homomorphism* rather than an *isomorphism*.

It is apparent that in Section 1, we showed that the binary structures $\langle U, \cdot \rangle$ and $\langle \mathbb{R}_c, +_c \rangle$ are isomorphic for all $c \in \mathbb{R}^+$. Also, $\langle U_n, \cdot \rangle$ and $\langle \mathbb{Z}_n, +_n \rangle$ are isomorphic for each $n \in \mathbb{Z}^+$.

Exercise 27 asks us to show that for a collection of binary algebraic structures, the relation $\simeq$ in Definition 3.7 is an equivalence relation on the collection. Our discussion leading to the preceding definition shows that the binary structures defined by Tables 3.1 through 3.4 are in the same equivalence class, while those given by Tables 3.5 and 3.6 are in different equivalence classes. We proceed to discuss how to try to determine whether binary structures are isomorphic.

How to Show That Binary Structures Are Isomorphic

We now give an outline showing how to proceed from Definition 3.7 to show that two binary structures $\langle S, * \rangle$ and $\langle S', *' \rangle$ are isomorphic.

Step 1 *Define the function ϕ that gives the isomorphism of S with S'.* Now this means that we have to describe, in some fashion, what $\phi(s)$ is to be for every $s \in S$.

Step 2 *Show that ϕ is a one-to-one function.* That is, suppose that $\phi(x) = \phi(y)$ in S' and deduce from this that $x = y$ in S.

Step 3 *Show that ϕ is onto S'.* That is, suppose that $s' \in S'$ is given and show that there does exist $s \in S$ such that $\phi(s) = s'$.

Step 4 *Show that $\phi(x * y) = \phi(x) *' \phi(y)$ for all $x, y \in S$.* This is just a question of computation. Compute both sides of the equation and see whether they are the same.

3.8 Example Let us show that the binary structure $\langle \mathbb{R}, + \rangle$ with operation the usual addition is isomorphic to the structure $\langle \mathbb{R}^+, \cdot \rangle$ where $\cdot$ is the usual multiplication.

Step 1 We have to somehow convert an operation of addition to multiplication. Recall from $a^{b+c} = (a^b)(a^c)$ that addition of exponents corresponds to multiplication of two quantities. Thus we try defining $\phi : \mathbb{R} \to \mathbb{R}^+$ by $\phi(x) = e^x$ for $x \in \mathbb{R}$. Note that $e^x > 0$ for all $x \in \mathbb{R}$, so indeed, $\phi(x) \in \mathbb{R}^+$.

Step 2 If $\phi(x) = \phi(y)$, then $e^x = e^y$. Taking the natural logarithm, we see that $x = y$, so ϕ is indeed one to one.

Step 3 If $r \in \mathbb{R}^+$, then $\ln(r) \in \mathbb{R}$ and $\phi(\ln r) = e^{\ln r} = r$. Thus ϕ is onto $\mathbb{R}^+$.

Step 4 For $x, y \in \mathbb{R}$, we have $\phi(x + y) = e^{x+y} = e^x \cdot e^y = \phi(x) \cdot \phi(y)$. Thus we see that ϕ is indeed an isomorphism. ▲

3.9 Example Let $2\mathbb{Z} = \{2n \mid n \in \mathbb{Z}\}$, so that $2\mathbb{Z}$ is the set of all even integers, positive, negative, and zero. We claim that $\langle \mathbb{Z}, + \rangle$ is isomorphic to $\langle 2\mathbb{Z}, + \rangle$, where $+$ is the usual addition. This will give an example of a binary structure $\langle \mathbb{Z}, + \rangle$ that is actually isomorphic to a structure consisting of a proper subset under the *induced* operation, in contrast to Example 3.8, where the operations were totally different.

Step 1 The obvious function $\phi : \mathbb{Z} \to 2\mathbb{Z}$ to try is given by $\phi(n) = 2n$ for $n \in \mathbb{Z}$.

Step 2 If $\phi(m) = \phi(n)$, then $2m = 2n$ so $m = n$. Thus ϕ is one to one.

Step 3 If $n \in 2\mathbb{Z}$, then n is even so $n = 2m$ for $m = n/2 \in \mathbb{Z}$. Hence $\phi(m) = 2(n/2) = n$ so ϕ is onto $2\mathbb{Z}$.

Step 4 Let $m, n \in \mathbb{Z}$. The equation

$$\phi(m + n) = 2(m + n) = 2m + 2n = \phi(m) + \phi(n)$$

then shows that ϕ is an isomorphism. ▲

How to Show That Binary Structures Are Not Isomorphic

We now turn to the reverse question, namely:

> *How do we demonstrate that two binary structures $\langle S, * \rangle$ and $\langle S', *' \rangle$ are not isomorphic, if this is the case?*

This would mean that there is no one-to-one function ϕ from S onto S' with the property $\phi(x * y) = \phi(x) *' \phi(y)$ for all $x, y \in S$. In general, it is clearly not feasible to try every possible one-to-one function mapping S onto S' and test whether it has this property, except in the case where there are *no* such functions. This is the case precisely when S and S' do not have the same cardinality. (See Definition 0.13.)

3.10 Example The binary structures $\langle \mathbb{Q}, + \rangle$ and $\langle \mathbb{R}, + \rangle$ are not isomorphic because $\mathbb{Q}$ has cardinality $\aleph_0$ while $|\mathbb{R}| \neq \aleph_0$. (See the discussion following Example 0.13.) Note that it is not enough to say that $\mathbb{Q}$ is a proper subset of $\mathbb{R}$. Example 3.9 shows that a proper subset with the induced operation can indeed be isomorphic to the entire binary structure. ▲

A **structural property** of a binary structure is one that must be shared by any isomorphic structure. It is not concerned with names or some other nonstructural characteristics of the elements. For example, the binary structures defined by Tables 3.1 and 3.2 are isomorphic, although the elements are totally different. Also, a structural property is not concerned with what we consider to be the "name" of the binary operation. Example 3.8 showed that a binary structure whose operation is our usual addition can be isomorphic to one whose operation is our usual multiplication. The number of elements in the set S is a structural property of $\langle S, * \rangle$.

In the event that there are one-to-one mappings of S onto S', we usually show that $\langle S, * \rangle$ is not isomorphic to $\langle S', *' \rangle$ (if this is the case) by showing that one has some structural property that the other does not possess.

3.11 Example The sets $\mathbb{Z}$ and $\mathbb{Z}^+$ both have cardinality $\aleph_0$, and there are lots of one-to-one functions mapping $\mathbb{Z}$ onto $\mathbb{Z}^+$. However, the binary structures $\langle \mathbb{Z}, \cdot \rangle$ and $\langle \mathbb{Z}^+, \cdot \rangle$, where $\cdot$ is the usual multiplication, are not isomorphic. In $\langle \mathbb{Z}, \cdot \rangle$ there are two elements x such that $x \cdot x = x$, namely, 0 and 1. However, in $\langle \mathbb{Z}^+, \cdot \rangle$, there is only the single element 1. ▲

We list a few examples of possible structural properties and nonstructural properties of a binary structure $\langle S, * \rangle$ to get you thinking along the right line.

Possible Structural Properties	*Possible Nonstructural Properties*
1. The set has 4 elements.	a. The number 4 is an element.
2. The operation is commutative.	b. The operation is called "addition."
3. $x * x = x$ for all $x \in S$.	c. The elements of S are matrices.
4. The equation $a * x = b$ has a solution x in S for all $a, b \in S$.	d. S is a subset of $\mathbb{C}$.

We introduced the algebraic notions of commutativity and associativity in Section 2. One other structural notion that will be of interest to us is illustrated by Table 3.3, where for the binary operation $*''$ on the set $\{x, y, z\}$, we have $x *'' u = u *'' x = u$ for all choices possible choices, x, y, and z for u. Thus x plays the same role as 0 in $\langle \mathbb{R}, + \rangle$ where $0 + u = u + 0 = u$ for all $u \in \mathbb{R}$, and the same role as 1 in $\langle \mathbb{R}, \cdot \rangle$ where $1 \cdot u = u \cdot 1 = u$ for all $u \in \mathbb{R}$. Because Tables 3.1 and 3.2 give structures isomorphic to the one in Table 3.3, they must exhibit an element with a similar property. We see that $b * u = u * b = u$ for all elements u appearing in Table 3.1 and that $\$ *' u = u *' \$ = u$ for all elements u in Table 3.2. We give a formal definition of this structural notion and prove a little theorem.

3.12 Definition Let $\langle S, * \rangle$ be a binary structure. An element e of S is an **identity element for** $*$ if $e * s = s * e = s$ for all $s \in S$. ■

3.13 Theorem **(Uniqueness of Identity Element)** A binary structure $\langle S, * \rangle$ has at most one identity element. That is, if there is an identity element, it is unique.

Proof Proceeding in the standard way to show uniqueness, suppose that both e and $\bar{e}$ are elements of S serving as identity elements. We let them compete with each other. Regarding e as an identity element, we must have $e * \bar{e} = \bar{e}$. However, regarding $\bar{e}$ as an identity element, we must have $e * \bar{e} = e$. We thus obtain $e = \bar{e}$, showing that an identity element must be unique. ◆

If you now have a good grasp of the notion of isomorphic binary structures, it should be evident that having an identity element for $*$ is indeed a structural property of a structure $\langle S, * \rangle$. However, we know from experience that many readers will be unable to see the forest because of all the trees that have appeared. For them, we now supply a careful proof, skipping along to touch those trees that are involved.

3.14 Theorem Suppose $\langle S, * \rangle$ has an identity element e for $*$. If $\phi : S \to S'$ is an isomorphism of $\langle S, * \rangle$ with $\langle S', *' \rangle$, then $\phi(e)$ is an identity element for the binary operation $*'$ on S'.

Proof Let $s' \in S'$. We must show that $\phi(e) *' s' = s' *' \phi(e) = s'$. Because ϕ is an isomorphism, it is a one-to-one map of S onto S'. In particular, there exists $s \in S$ such that $\phi(s) = s'$. Now e is an identity element for $*$ so that we know that $e * s = s * e = s$. Because ϕ is a function, we then obtain

$$\phi(e * s) = \phi(s * e) = \phi(s).$$

Using Definition 3.7 of an isomorphism, we can rewrite this as

$$\phi(e) *' \phi(s) = \phi(s) *' \phi(e) = \phi(s).$$

Remembering that we chose $s \in S$ such that $\phi(s) = s'$, we obtain the desired relation $\phi(e) *' s' = s' *' \phi(e) = s'$. ◆

We conclude with three more examples showing via structural properties that certain binary structures are not isomorphic. In the exercises we ask you to show, as in Theorem 3.14, that the properties we use to distinguish the structures in these examples are indeed structural. That is, they must be shared by any isomorphic structure.

3.15 Example We show that the binary structures $\langle \mathbb{Q}, + \rangle$ and $\langle \mathbb{Z}, + \rangle$ under the usual addition are not isomorphic. (Both $\mathbb{Q}$ and $\mathbb{Z}$ have cardinality $\aleph_0$, so there are lots of one-to-one functions mapping $\mathbb{Q}$ onto $\mathbb{Z}$.) The equation $x + x = c$ has a solution x for all $c \in \mathbb{Q}$, but this is not the case in $\mathbb{Z}$. For example, the equation $x + x = 3$ has no solution in $\mathbb{Z}$. We have exhibited a structural property that *distinguishes* these two structures. ▲

3.16 Example The binary structures $\langle \mathbb{C}, \cdot \rangle$ and $\langle \mathbb{R}, \cdot \rangle$ under the usual multiplication are not isomorphic. (It can be shown that $\mathbb{C}$ and $\mathbb{R}$ have the same cardinality.) The equation $x \cdot x = c$ has a solution x for all $c \in \mathbb{C}$, but $x \cdot x = -1$ has no solution in $\mathbb{R}$. ▲

3.17 Example The binary structure $\langle M_2(\mathbb{R}), \cdot \rangle$ of 2×2 real matrices with the usual matrix multiplication is not isomorphic to $\langle \mathbb{R}, \cdot \rangle$ with the usual number multiplication. (It can be shown that both sets have cardinality $|\mathbb{R}|$.) Multiplication of numbers is commutative, but multiplication of matrices is not. ▲

EXERCISES 3

In all the exercises, $+$ is the usual addition on the set where it is specified, and $\cdot$ is the usual multiplication.

Computations

1. What three things must we check to determine whether a function $\phi: S \to S'$ is an isomorphism of a binary structure $\langle S, * \rangle$ with $\langle S', *' \rangle$?

In Exercises 2 through 10, determine whether the given map ϕ is an isomorphism of the first binary structure with the second. (See Exercise 1.) If it is not an isomorphism, why not?

2. $\langle \mathbb{Z}, + \rangle$ with $\langle \mathbb{Z}, + \rangle$ where $\phi(n) = -n$ for $n \in \mathbb{Z}$
3. $\langle \mathbb{Z}, + \rangle$ with $\langle \mathbb{Z}, + \rangle$ where $\phi(n) = 2n$ for $n \in \mathbb{Z}$
4. $\langle \mathbb{Z}, + \rangle$ with $\langle \mathbb{Z}, + \rangle$ where $\phi(n) = n + 1$ for $n \in \mathbb{Z}$
5. $\langle \mathbb{Q}, + \rangle$ with $\langle \mathbb{Q}, + \rangle$ where $\phi(x) = x/2$ for $x \in \mathbb{Q}$
6. $\langle \mathbb{Q}, \cdot \rangle$ with $\langle \mathbb{Q}, \cdot \rangle$ where $\phi(x) = x^2$ for $x \in \mathbb{Q}$
7. $\langle \mathbb{R}, \cdot \rangle$ with $\langle \mathbb{R}, \cdot \rangle$ where $\phi(x) = x^3$ for $x \in \mathbb{R}$
8. $\langle M_2(\mathbb{R}), \cdot \rangle$ with $\langle \mathbb{R}, \cdot \rangle$ where $\phi(A)$ is the determinant of matrix A
9. $\langle M_1(\mathbb{R}), \cdot \rangle$ with $\langle \mathbb{R}, \cdot \rangle$ where $\phi(A)$ is the determinant of matrix A
10. $\langle \mathbb{R}, + \rangle$ with $\langle \mathbb{R}^+, \cdot \rangle$ where $\phi(r) = 0.5^r$ for $r \in \mathbb{R}$

In Exercises 11 through 15, let F be the set of all functions f mapping $\mathbb{R}$ into $\mathbb{R}$ that have derivatives of all orders. Follow the instructions for Exercises 2 through 10.

11. $\langle F, + \rangle$ with $\langle F, + \rangle$ where $\phi(f) = f'$, the derivative of f
12. $\langle F, + \rangle$ with $\langle \mathbb{R}, + \rangle$ where $\phi(f) = f'(0)$
13. $\langle F, + \rangle$ with $\langle F, + \rangle$ where $\phi(f)(x) = \int_0^x f(t)dt$
14. $\langle F, + \rangle$ with $\langle F, + \rangle$ where $\phi(f)(x) = \frac{d}{dx}[\int_0^x f(t)dt]$
15. $\langle F, \cdot \rangle$ with $\langle F, \cdot \rangle$ where $\phi(f)(x) = x \cdot f(x)$
16. The map $\phi : \mathbb{Z} \to \mathbb{Z}$ defined by $\phi(n) = n + 1$ for $n \in \mathbb{Z}$ is one to one and onto $\mathbb{Z}$. Give the definition of a binary operation $*$ on $\mathbb{Z}$ such that ϕ is an isomorphism mapping

    a. $\langle \mathbb{Z}, + \rangle$ onto $\langle \mathbb{Z}, * \rangle$, **b.** $\langle \mathbb{Z}, * \rangle$ onto $\langle \mathbb{Z}, + \rangle$.

    In each case, give the identity element for $*$ on $\mathbb{Z}$.

17. The map $\phi : \mathbb{Z} \to \mathbb{Z}$ defined by $\phi(n) = n + 1$ for $n \in \mathbb{Z}$ is one to one and onto $\mathbb{Z}$. Give the definition of a binary operation $*$ on $\mathbb{Z}$ such that ϕ is an isomorphism mapping

    a. $\langle \mathbb{Z}, \cdot \rangle$ onto $\langle \mathbb{Z}, * \rangle$, **b.** $\langle \mathbb{Z}, * \rangle$ onto $\langle \mathbb{Z}, \cdot \rangle$.

    In each case, give the identity element for $*$ on $\mathbb{Z}$.

18. The map $\phi : \mathbb{Q} \to \mathbb{Q}$ defined by $\phi(x) = 3x - 1$ for $x \in \mathbb{Q}$ is one to one and onto $\mathbb{Q}$. Give the definition of a binary operation $*$ on $\mathbb{Q}$ such that ϕ is an isomorphism mapping

    a. $\langle \mathbb{Q}, + \rangle$ onto $\langle \mathbb{Q}, * \rangle$, **b.** $\langle \mathbb{Q}, * \rangle$ onto $\langle \mathbb{Q}, + \rangle$.

    In each case, give the identity element for $*$ on $\mathbb{Q}$.

19. The map $\phi : \mathbb{Q} \to \mathbb{Q}$ defined by $\phi(x) = 3x - 1$ for $x \in \mathbb{Q}$ is one to one and onto $\mathbb{Q}$. Give the definition of a binary operation $*$ on $\mathbb{Q}$ such that ϕ is an isomorphism mapping

a. $\langle \mathbb{Q}, \cdot \rangle$ onto $\langle \mathbb{Q}, * \rangle$, **b.** $\langle \mathbb{Q}, * \rangle$ onto $\langle \mathbb{Q}, \cdot \rangle$.

In each case, give the identity element for $*$ on $\mathbb{Q}$.

Concepts

20. The displayed homomorphism condition for an isomorphism ϕ in Definition 3.7 is sometimes summarized by saying, "ϕ must commute with the binary operation(s)." Explain how that condition can be viewed in this manner.

In Exercises 21 and 22, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

21. A function $\phi : S \to S'$ is an *isomorphism* if and only if $\phi(a * b) = \phi(a) *' \phi(b)$.

22. Let $*$ be a binary operation on a set S. An element e of S with the property $s * e = s = e * s$ is an *identity element for* $*$ for all $s \in S$.

Proof Synopsis

A good test of your understanding of a proof is your ability to give a one or two sentence synopsis of it, explaining the idea of the proof without all the details and computations. Note that we said "sentence" and not "equation." From now on, some of our exercise sets may contain one or two problems asking for a synopsis of a proof in the text. It should rarely exceed three sentences. We should illustrate for you what we mean by a synopsis. Here is our one-sentence synopsis of Theorem 3.14. Read the statement of the theorem now, and then our synopsis.

> Representing an element of S' as $\phi(s)$ for some $s \in S$, use the homomorphism property of ϕ to carry the computation of $\phi(e) *' \phi(s)$ back to a computation in S.

That is the kind of explanation that one mathematician might give another if asked, "How does the proof go?" We did not make the computation or explain why we could represent an element of S' as $\phi(s)$. To supply every detail would result in a completely written proof. We just gave the guts of the argument in our synopsis.

23. Give a proof synopsis of Theorem 3.13.

Theory

24. An identity element for a binary operation $*$ as described by Definition 3.12 is sometimes referred to as "a two-sided identity element." Using complete sentences, give analogous definitions for

a. a *left identity element* e_L for $*$, and **b.** a *right identity element* e_R for $*$.

Theorem 3.13 shows that if a two-sided identity element for $*$ exists, it is unique. Is the same true for a one-sided identity element you just defined? If so, prove it. If not, give a counterexample $\langle S, * \rangle$ for a finite set S and find the first place where the proof of Theorem 3.13 breaks down.

25. Continuing the ideas of Exercise 24 can a binary structure have a left identity element e_L and a right identity element e_R where $e_L \neq e_R$? If so, give an example, using an operation on a finite set S. If not, prove that it is impossible.

26. Recall that if $f : A \to B$ is a one-to-one function mapping A onto B, then $f^{-1}(b)$ is the unique $a \in A$ such that $f(a) = b$. Prove that if $\phi : S \to S'$ is an isomorphism of $\langle S, * \rangle$ with $\langle S', *' \rangle$, then ϕ^{-1} is an isomorphism of $\langle S', *' \rangle$ with $\langle S, * \rangle$.

27. Prove that if $\phi : S \to S'$ is an isomorphism of $\langle S, * \rangle$ with $\langle S', *' \rangle$ and $\psi : S' \to S''$ is an isomorphism of $\langle S', *' \rangle$ with $\langle S'', *'' \rangle$, then the composite function $\psi \circ \phi$ is an isomorphism of $\langle S, * \rangle$ with $\langle S'', *'' \rangle$.

28. Prove that the relation $\simeq$ of being isomorphic, described in Definition 3.7, is an equivalence relation on any set of binary structures. You may simply quote the results you were asked to prove in the preceding two exercises at appropriate places in your proof.

In Exercises 29 through 32, give a careful proof for a skeptic that the indicated property of a binary structure $\langle S, * \rangle$ is indeed a structural property. (In Theorem 3.14, we did this for the property, "There is an identity element for $*$.")

29. The operation $*$ is commutative.

30. The operation $*$ is associative.

31. For each $c \in S$, the equation $x * x = c$ has a solution x in S.

32. There exists an element b in S such that $b * b = b$.

33. Let H be the subset of $M_2(\mathbb{R})$ consisting of all matrices of the form $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ for $a, b \in \mathbb{R}$. Exercise 23 of Section 2 shows that H is closed under both matrix addition and matrix multiplication.

a. Show that $\langle \mathbb{C}, + \rangle$ is isomorphic to $\langle H, + \rangle$.

b. Show that $\langle \mathbb{C}, \cdot \rangle$ is isomorphic to $\langle H, \cdot \rangle$.

(We say that H is a *matrix representation* of the complex numbers $\mathbb{C}$.)

34. There are 16 possible binary structures on the set $\{a, b\}$ of two elements. How many nonisomorphic (that is, structurally different) structures are there among these 16? Phrased more precisely in terms of the isomorphism equivalence relation $\simeq$ on this set of 16 structures, how many equivalence classes are there? Write down one structure from each equivalence class. [*Hint:* Interchanging a and b everywhere in a table and then rewriting the table with elements listed in the original order does not always yield a table different from the one we started with.]

SECTION 4 GROUPS

Let us continue the analysis of our past experience with algebra. Once we had mastered the computational problems of addition and multiplication of numbers, we were ready to apply these binary operations to the solution of problems. Often problems lead to equations involving some unknown number x, which is to be determined. The simplest equations are the linear ones of the forms $a + x = b$ for the operation of addition, and $ax = b$ for multiplication. The additive linear equation always has a numerical solution, and so has the multiplicative one, provided $a \neq 0$. Indeed, the need for solutions of additive linear equations such as $5 + x = 2$ is a very good motivation for the negative numbers. Similarly, the need for rational numbers is shown by equations such as $2x = 3$.

It is desirable for us to be able to solve linear equations involving our binary operations. This is not possible for every binary operation, however. For example, the equation $a * x = a$ has no solution in $S = \{a, b, c\}$ for the operation $*$ of Example 2.14. Let us abstract from familiar algebra those properties of addition that enable us to solve the equation $5 + x = 2$ in $\mathbb{Z}$. We must not refer to subtraction, for we are concerned with the solution phrased in terms of a single binary operation, in this case addition. The steps in

the solution are as follows:

$$\begin{aligned} 5+x &= 2, && \text{given,} \\ -5+(5+x) &= -5+2, && \text{adding } -5, \\ (-5+5)+x &= -5+2, && \text{associative law,} \\ 0+x &= -5+2, && \text{computing } -5+5, \\ x &= -5+2, && \text{property of 0,} \\ x &= -3, && \text{computing } -5+2. \end{aligned}$$

Strictly speaking, we have not shown here that -3 is a solution, but rather that it is the only possibility for a solution. To show that -3 is a solution, one merely computes $5+(-3)$. A similar analysis could be made for the equation $2x = 3$ in the rational numbers with the operation of multiplication:

$$\begin{aligned} 2x &= 3, && \text{given,} \\ \tfrac{1}{2}(2x) &= \tfrac{1}{2}(3), && \text{multiplying by } \tfrac{1}{2}, \\ (\tfrac{1}{2}\cdot 2)x &= \tfrac{1}{2}3, && \text{associative law,} \\ 1\cdot x &= \tfrac{1}{2}3, && \text{computing } \tfrac{1}{2}2, \\ x &= \tfrac{1}{2}3, && \text{property of 1,} \\ x &= \tfrac{3}{2}, && \text{computing } \tfrac{1}{2}3. \end{aligned}$$

We can now see what properties a set S and a binary operation $*$ on S would have to have to permit imitation of this procedure for an equation $a*x=b$ for $a, b \in S$. Basic to the procedure is the existence of an element e in S with the property that $e*x=x$ for all $x \in S$. For our additive example, 0 played the role of e, and 1 played the role for our multiplicative example. Then we need an element a' in S that has the property that $a'*a=e$. For our additive example with $a=5$, -5 played the role of a', and $\frac{1}{2}$ played the role for our multiplicative example with $a=2$. Finally we need the associative law. The remainder is just computation. A similar analysis shows that in order to solve the equation $x*a=b$ (remember that $a*x$ need not equal $x*a$), we would like to have an element e in S such that $x*e=x$ for all $x \in S$ and an a' in S such that $a*a'=e$. With all of these properties of $*$ on S, we could be sure of being able to solve linear equations. Thus we need an associative binary structure $\langle S, * \rangle$ with an identity element e such that for each $a \in S$, there exists $a' \in S$ such that $a*a'=a'*a=e$. This is precisely the notion of a *group*, which we now define.

Definition and Examples

Rather than describe a *group* using terms defined in Sections 2 and 3 as we did at the end of the preceding paragraph, we give a self-contained definition. This enables a person who picks up this text to discover what a group is without having to look up more terms.

4.1 Definition A **group** $\langle G, * \rangle$ is a set G, closed under a binary operation $*$, such that the following axioms are satisfied:

$\mathscr{G}_1$: For all $a, b, c \in G$, we have

$$(a*b)*c = a*(b*c). \quad \textbf{associativity of } *$$

$\mathscr{G}_2$: There is an element e in G such that for all $x \in G$,

$$e * x = x * e = x. \quad \textbf{identity element } e \text{ for } *$$

$\mathscr{G}_3$: Corresponding to each $a \in G$, there is an element a' in G such that

$$a * a' = a' * a = e. \quad \textbf{inverse } a' \text{ of } a$$ ■

4.2 Example We easily see that $\langle U, \cdot \rangle$ and $\langle U_n, \cdot \rangle$ are groups. Multiplication of complex numbers is associative and both U and U_n contain 1, which is an identity for multiplication. For $e^{i\theta} \in U$, the computation

$$e^{i\theta} \cdot e^{i(2\pi - \theta)} = e^{2\pi i} = 1$$

shows that every element of U has an inverse. For $z \in U_n$, the computation

$$z \cdot z^{n-1} = z^n = 1$$

shows that every element of U_n has an inverse. Thus $\langle U, \cdot \rangle$ and $\langle U_n, \cdot \rangle$ are groups. Because $\langle \mathbb{R}_c, +_c \rangle$ is isomorphic to $\langle U, \cdot \rangle$, we see that $\langle \mathbb{R}_c, +_c \rangle$ is a group for all $c \in \mathbb{R}^+$. Similarly, the fact that $\langle \mathbb{Z}_n, +_n \rangle$ is isomorphic to $\langle U_n, \cdot \rangle$ shows that $\langle \mathbb{Z}_n, +_n \rangle$ is a group for all $n \in \mathbb{Z}^+$. ▲

We point out now that we will sometimes be sloppy in notation. Rather than use the binary structure notation $\langle G, * \rangle$ constantly, we often refer to a group G, with the understanding that there is of course a binary operation on the set G. In the event that clarity demands that we specify an operation $*$ on G, we use the phrase "the group G

Historical Note

There are three historical roots of the development of abstract group theory evident in the mathematical literature of the nineteenth century: the theory of algebraic equations, number theory, and geometry. All three of these areas used group-theoretic methods of reasoning, although the methods were considerably more explicit in the first area than in the other two.

One of the central themes of geometry in the nineteenth century was the search for invariants under various types of geometric transformations. Gradually attention became focused on the transformations themselves, which in many cases can be thought of as elements of groups.

In number theory, already in the eighteenth century Leonhard Euler had considered the remainders on division of powers a^n by a fixed prime p. These remainders have "group" properties. Similarly, Carl F. Gauss, in his *Disquisitiones Arithmeticae* (1800), dealt extensively with quadratic forms $ax^2 + 2bxy + cy^2$, and in particular showed that equivalence classes of these forms under composition possessed what amounted to group properties.

Finally, the theory of algebraic equations provided the most explicit prefiguring of the group concept. Joseph-Louis Lagrange (1736–1813) in fact initiated the study of permutations of the roots of an equation as a tool for solving it. These permutations, of course, were ultimately considered as elements of a group.

It was Walter von Dyck (1856–1934) and Heinrich Weber (1842–1913) who in 1882 were able independently to combine the three historical roots and give clear definitions of the notion of an abstract group.

under $*$." For example, we may refer to the *groups* $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$ *under addition* rather than write the more tedious $\langle \mathbb{Z}, + \rangle$, $\langle \mathbb{Q}, + \rangle$, and $\langle \mathbb{R}, + \rangle$. However, we feel free to refer to the group $\mathbb{Z}_8$ without specifying the operation.

4.3 Definition A group G is **abelian** if its binary operation is commutative. ■

Historical Note

Commutative groups are called *abelian* in honor of the Norwegian mathematician Niels Henrik Abel (1802–1829). Abel was interested in the question of solvability of polynomial equations. In a paper written in 1828, he proved that if all the roots of such an equation can be expressed as rational functions $f, g, \ldots, h$ of one of them, say x, and if for any two of these roots, $f(x)$ and $g(x)$, the relation $f(g(x)) = g(f(x))$ always holds, then the equation is solvable by radicals. Abel showed that each of these functions in fact permutes the roots of the equation; hence, these functions are elements of the group of permutations of the roots. It was this property of commutativity in these permutation groups associated with solvable equations that led Camille Jordan in his 1870 treatise on algebra to name such groups *abelian;* the name since then has been applied to commutative groups in general.

Abel was attracted to mathematics as a teenager and soon surpassed all his teachers in Norway. He finally received a government travel grant to study elsewhere in 1825 and proceeded to Berlin, where he befriended August Crelle, the founder of the most influential German mathematical journal. Abel contributed numerous papers to Crelle's *Journal* during the next several years, including many in the field of elliptic functions, whose theory he created virtually single-handedly. Abel returned to Norway in 1827 with no position and an abundance of debts. He nevertheless continued to write brilliant papers, but died of tuberculosis at the age of 26, two days before Crelle succeeded in finding a university position for him in Berlin.

Let us give some examples of some sets with binary operations that give groups and also of some that do not give groups.

4.4 Example The set $\mathbb{Z}^+$ under addition is *not* a group. There is no identity element for $+$ in $\mathbb{Z}^+$. ▲

4.5 Example The set of all nonnegative integers (including 0) under addition is still *not* a group. There is an identity element 0, but no inverse for 2. ▲

4.6 Example The familiar additive properties of integers and of rational, real, and complex numbers show that $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ under addition are abelian groups. ▲

4.7 Example The set $\mathbb{Z}^+$ under multiplication is *not* a group. There is an identity 1, but no inverse of 3. ▲

4.8 Example The familiar multiplicative properties of rational, real, and complex numbers show that the sets $\mathbb{Q}^+$ and $\mathbb{R}^+$ of positive numbers and the sets $\mathbb{Q}^*$, $\mathbb{R}^*$, and $\mathbb{C}^*$ of nonzero numbers under multiplication are abelian groups. ▲

4.9 Example The set of all real-valued functions with domain $\mathbb{R}$ under function addition is a group. This group is abelian. ▲

4.10 Example **(Linear Algebra)** Those who have studied vector spaces should note that the axioms for a vector space V pertaining just to vector addition can be summarized by asserting that V under vector addition is an abelian group. ▲

4.11 Example The set $M_{m\times n}(\mathbb{R})$ of all $m \times n$ matrices under matrix addition is a group. The $m \times n$ matrix with all entries 0 is the identity matrix. This group is abelian. ▲

4.12 Example The set $M_n(\mathbb{R})$ of all $n \times n$ matrices under matrix multiplication is *not* a group. The $n \times n$ matrix with all entries 0 has no inverse. ▲

4.13 Example Show that the subset S of $M_n(\mathbb{R})$ consisting of all *invertible* $n \times n$ matrices under matrix multiplication is a group.

Solution We start by showing that S is closed under matrix multiplication. Let A and B be in S, so that both A^{-1} and B^{-1} exist and $AA^{-1} = BB^{-1} = I_n$. Then

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AI_nA^{-1} = I_n,$$

so that AB is invertible and consequently is also in S.

Since matrix multiplication is associative and I_n acts as the identity element, and since each element of S has an inverse by definition of S, we see that S is indeed a group. This group is *not* commutative. It is our first example of a *nonabelian group.* ▲

The group of invertible $n \times n$ matrices described in the preceding example is of fundamental importance in linear algebra. It is the **general linear group of degree** n, and is usually denoted by $GL(n, \mathbb{R})$. Those of you who have studied linear algebra know that a matrix A in $GL(n, \mathbb{R})$ gives rise to an invertible linear transformation $T : \mathbb{R}^n \to \mathbb{R}^n$, defined by $T(\mathbf{x}) = A\mathbf{x}$, and that conversely, every invertible linear transformation of $\mathbb{R}^n$ into itself is defined in this fashion by some matrix in $GL(n, \mathbb{R})$. Also, matrix multiplication corresponds to composition of linear transformations. Thus all invertible linear transformations of $\mathbb{R}^n$ into itself form a group under function composition; this group is usually denoted by $GL(\mathbb{R}^n)$. Of course, $GL(n, \mathbb{R}) \simeq GL(\mathbb{R}^n)$.

4.14 Example Let $*$ be defined on $\mathbb{Q}^+$ by $a * b = ab/2$. Then

$$(a * b) * c = \frac{ab}{2} * c = \frac{abc}{4},$$

and likewise

$$a * (b * c) = a * \frac{bc}{2} = \frac{abc}{4}.$$

Thus $*$ is associative. Computation shows that

$$2 * a = a * 2 = a$$

for all $a \in \mathbb{Q}^+$, so 2 is an identity element for $*$. Finally,

$$a * \frac{4}{a} = \frac{4}{a} * a = 2,$$

so $a' = 4/a$ is an inverse for a. Hence $\mathbb{Q}^+$ with the operation $*$ is a group. ▲

Elementary Properties of Groups

As we proceed to prove our first theorem about groups, we must use Definition 4.1, which is the only thing we know about groups at the moment. The proof of a second theorem can employ both Definition 4.1 and the first theorem; the proof of a third theorem can use the definition and the first two theorems, and so on.

Our first theorem will establish cancellation laws. In real arithmetic, we know that $2a = 2b$ implies that $a = b$. We need only divide both sides of the equation $2a = 2b$ by 2, or equivalently, multiply both sides by $\frac{1}{2}$, which is the multiplicative inverse of 2. We parrot this proof to establish cancellation laws for any group. Note that we will also use the associative law.

4.15 Theorem If G is a group with binary operation $*$, then the **left and right cancellation laws** hold in G, that is, $a * b = a * c$ implies $b = c$, and $b * a = c * a$ implies $b = c$ for all $a, b, c \in G$.

Proof Suppose $a * b = a * c$. Then by $\mathscr{G}_3$, there exists a', and

$$a' * (a * b) = a' * (a * c).$$

By the associative law,

$$(a' * a) * b = (a' * a) * c.$$

By the definition of a' in $\mathscr{G}_3$, $a' * a = e$, so

$$e * b = e * c.$$

By the definition of e in $\mathscr{G}_2$,

$$b = c.$$

Similarly, from $b * a = c * a$ one can deduce that $b = c$ upon multiplication on the right by a' and use of the axioms for a group. ◆

Our next proof can make use of Theorem 4.15. We show that a "linear equation" in a group has a *unique* solution. Recall that we chose our group properties to allow us to find solutions of such equations.

4.16 Theorem If G is a group with binary operation $*$, and if a and b are any elements of G, then the linear equations $a * x = b$ and $y * a = b$ have unique solutions x and y in G.

Proof First we show the existence of *at least* one solution by just computing that $a' * b$ is a solution of $a * x = b$. Note that

$$\begin{aligned} a * (a' * b) &= (a * a') * b, && \text{associative law,} \\ &= e * b, && \text{definition of } a', \\ &= b, && \text{property of } e. \end{aligned}$$

Thus $x = a' * b$ is a solution of $a * x = b$. In a similar fashion, $y = b * a'$ is a solution of $y * a = b$.

To show uniqueness of y, we use the standard method of assuming that we have two solutions, y_1 and y_2, so that $y_1 * a = b$ and $y_2 * a = b$. Then $y_1 * a = y_2 * a$, and by Theorem 4.15, $y_1 = y_2$. The uniqueness of x follows similarly. ◆

Of course, to prove the uniqueness in the last theorem, we could have followed the procedure we used in motivating the definition of a group, showing that if $a * x = b$, then $x = a' * b$. However, we chose to illustrate the standard way to prove an object is unique; namely, suppose you have two such objects, and then prove they must be the same. Note that the solutions $x = a' * b$ and $y = b * a'$ need not be the same unless $*$ is commutative.

Because a group is a special type of binary structure, we know from Theorem 3.13 that the identity e in a group is unique. We state this again as part of the next theorem for easy reference.

4.17 Theorem In a group G with binary operation $*$, there is only one element e in G such that

$$e * x = x * e = x$$

for all $x \in G$. Likewise for each $a \in G$, there is only one element a' in G such that

$$a' * a = a * a' = e.$$

In summary, the identity element and inverse of each element are unique in a group.

Proof Theorem 3.13 shows that an identity element for any binary structure is unique. No use of the group axioms was required to show this.

Turning to the uniqueness of an inverse, suppose that $a \in G$ has inverses a' and a'' so that $a' * a = a * a' = e$ and $a'' * a = a * a'' = e$. Then

$$a * a'' = a * a' = e$$

and, by Theorem 4.15,

$$a'' = a',$$

so the inverse of a in a group is unique. ◆

Note that in a group G, we have

$$(a * b) * (b' * a') = a * (b * b') * a' = (a * e) * a' = a * a' = e.$$

This equation and Theorem 4.17 show that $b' * a'$ is the unique inverse of $a * b$. That is, $(a * b)' = b' * a'$. We state this as a corollary.

4.18 Corollary Let G be a group. For all $a, b \in G$, we have $(a * b)' = b' * a'$.

For your information, we remark that binary algebraic structures with weaker axioms than those for a group have also been studied quite extensively. Of these weaker structures, the **semigroup,** a set with an associative binary operation, has perhaps had the most attention. A **monoid** is a semigroup that has an identity element for the binary operation. Note that every group is both a semigroup and a monoid.

Finally, it is possible to give axioms for a group $\langle G, * \rangle$ that seem at first glance to be weaker, namely:

1. The binary operation $*$ on G is associative.
2. There exists a **left identity element** e in G such that $e * x = x$ for all $x \in G$.
3. For each $a \in G$, there exists **a left inverse** a' in G such that $a' * a = e$.

From this *one-sided definition*, one can prove that the left identity element is also a right identity element, and a left inverse is also a right inverse for the same element. Thus these axioms should not be called *weaker*, since they result in exactly the same structures being called groups. It is conceivable that it might be easier in some cases to check these *left axioms* than to check our *two-sided axioms*. Of course, by symmetry it is clear that there are also *right axioms* for a group.

Finite Groups and Group Tables

All our examples after Example 4.2 have been of infinite groups, that is, groups where the set G has an infinite number of elements. We turn to finite groups, starting with the smallest finite sets.

Since a group has to have at least one element, namely, the identity, a minimal set that might give rise to a group is a one-element set $\{e\}$. The only possible binary operation $*$ on $\{e\}$ is defined by $e * e = e$. The three group axioms hold. The identity element is always its own inverse in every group.

Let us try to put a group structure on a set of two elements. Since one of the elements must play the role of identity element, we may as well let the set be $\{e, a\}$. Let us attempt to find a table for a binary operation $*$ on $\{e, a\}$ that gives a group structure on $\{e, a\}$. When giving a table for a group operation, we shall always list the identity first, as in the following table.

$*$	e	a
e		
a		

Since e is to be the identity, so

$$e * x = x * e = x$$

for all $x \in \{e, a\}$, we are forced to fill in the table as follows, if $*$ is to give a group:

$*$	e	a
e	e	a
a	a	

Also, a must have an inverse a' such that

$$a * a' = a' * a = e.$$

In our case, a' must be either e or a. Since $a' = e$ obviously does not work, we must have $a' = a$, so we have to complete the table as follows:

$*$	e	a
e	e	a
a	a	e

All the group axioms are now satisfied, except possibly the associative property. Checking associativity on a case-by-case basis from a table defining an operation can be a very tedious process. However, we know that $\mathbb{Z}_2 = \{0, 1\}$ under addition modulo 2 is a group, and by our arguments, its table must be the one above with e replaced by 0 and a by 1. Thus the associative property must be satisfied for our table containing e and a.

With this example as background, we should be able to list some necessary conditions that a table giving a binary operation on a finite set must satisfy for the operation to give a group structure on the set. There must be one element of the set, which we may as well denote by e, that acts as the identity element. The condition $e * x = x$ means that the row of the table opposite e at the extreme left must contain exactly the elements appearing across the very top of the table in the same order. Similarly, the condition $x * e = x$ means that the column of the table under e at the very top must contain exactly the elements appearing at the extreme left in the same order. The fact that every element a has a right and a left inverse means that in the row having a at the extreme left, the element e must appear, and in the column under a at the very top, the e must appear. Thus e must appear in each row and in each column. We can do even better than this, however. By Theorem 4.16, not only the equations $a * x = e$ and $y * a = e$ have unique solutions, but also the equations $a * x = b$ and $y * a = b$. By a similar argument, this means that *each element b of the group must appear once and only once in each row and each column of the table.*

Suppose conversely that a table for a binary operation on a finite set is such that there is an element acting as identity and that in each row and each column, each element of the set appears exactly once. Then it can be seen that the structure is a group structure if and only if the associative law holds. If a binary operation $*$ is given by a table, the associative law is usually messy to check. If the operation $*$ is defined by some characterizing property of $a * b$, the associative law is often easy to check. Fortunately, this second case turns out to be the one usually encountered.

We saw that there was essentially only one group of two elements in the sense that if the elements are denoted by e and a with the identity element e appearing first, the table must be shown in Table 4.19. Suppose that a set has three elements. As before, we may as well let the set be $\{e, a, b\}$. For e to be an identity element, a binary operation $*$ on this set has to have a table of the form shown in Table 4.20. This leaves four places to be filled in. You can quickly see that Table 4.20 must be completed as shown in Table 4.21 if each row and each column are to contain each element exactly once. Because there was only one way to complete the table and $\mathbb{Z}_3 = \{0, 1, 2\}$ under addition modulo 3 is a group, the associative property must hold for our table containing e, a, and b.

Now suppose that G' is any other group of three elements and imagine a table for G' with identity element appearing first. Since our filling out of the table for $G = \{e, a, b\}$ could be done in only one way, we see that if we take the table for G' and rename the identity e, the next element listed a, and the last element b, the resulting table for G' must be the same as the one we had for G. As explained in Section 3, this renaming gives an isomorphism of the group G' with the group G. Definition 3.7 defined the notion of *isomorphism* and of *isomorphic binary structures*. Groups are just certain types of binary structures, so the same definition pertains to them. Thus our work above can be summarized by saying that all groups with a single element are isomorphic, all groups with just two elements are isomorphic, and all groups with just three elements are isomorphic. We use the phrase *up to isomorphism* to express this identification using the equivalence relation $\simeq$. Thus we may say, "There is only one group of three elements, up to isomorphism."

4.19 Table

$*$	e	a
e	e	a
a	a	e

4.20 Table

$*$	e	a	b
e	e	a	b
a	a		
b	b		

4.21 Table

$*$	e	a	b
e	e	a	b
a	a	b	e
b	b	e	a

EXERCISES 4

Computations

In Exercises 1 through 6, determine whether the binary operation $*$ gives a group structure on the given set. If no group results, give the first axiom in the order $\mathscr{G}_1$, $\mathscr{G}_2$, $\mathscr{G}_3$ from Definition 4.1 that does not hold.

1. Let $*$ be defined on $\mathbb{Z}$ by letting $a * b = ab$.
2. Let $*$ be defined on $2\mathbb{Z} = \{2n \mid n \in \mathbb{Z}\}$ by letting $a * b = a + b$.
3. Let $*$ be defined on $\mathbb{R}^+$ by letting $a * b = \sqrt{ab}$.
4. Let $*$ be defined on $\mathbb{Q}$ by letting $a * b = ab$.
5. Let $*$ be defined on the set $\mathbb{R}^*$ of nonzero real numbers by letting $a * b = a/b$.
6. Let $*$ be defined on $\mathbb{C}$ by letting $a * b = |ab|$.
7. Give an example of an abelian group G where G has exactly 1000 elements.
8. We can also consider multiplication $\cdot_n$ modulo n in $\mathbb{Z}_n$. For example, $5 \cdot_7 6 = 2$ in $\mathbb{Z}_7$ because $5 \cdot 6 = 30 = 4(7) + 2$. The set $\{1, 3, 5, 7\}$ with multiplication $\cdot_8$ modulo 8 is a group. Give the table for this group.
9. Show that the group $\langle U, \cdot \rangle$ is not isomorphic to either $\langle \mathbb{R}, + \rangle$ or $\langle \mathbb{R}^*, \cdot \rangle$. (All three groups have cardinality $|\mathbb{R}|$.)
10. Let n be a positive integer and let $n\mathbb{Z} = \{nm \mid m \in \mathbb{Z}\}$.
    a. Show that $\langle n\mathbb{Z}, + \rangle$ is a group.
    b. Show that $\langle n\mathbb{Z}, + \rangle \simeq \langle \mathbb{Z}, + \rangle$.

In Exercises 11 through 18, determine whether the given set of matrices under the specified operation, matrix addition or multiplication, is a group. Recall that a **diagonal matrix** is a square matrix whose only nonzero entries lie on the **main diagonal,** from the upper left to the lower right corner. An **upper-triangular matrix** is a square matrix with only zero entries below the main diagonal. Associated with each $n \times n$ matrix A is a number called the determinant of A, denoted by $\det(A)$. If A and B are both $n \times n$ matrices, then $\det(AB) = \det(A)\det(B)$. Also, $\det(I_n) = 1$ and A is invertible if and only if $\det(A) \neq 0$.

11. All $n \times n$ diagonal matrices under matrix addition.

12. All $n \times n$ diagonal matrices under matrix multiplication.

13. All $n \times n$ diagonal matrices with no zero diagonal entry under matrix multiplication.

14. All $n \times n$ diagonal matrices with all diagonal entries 1 or -1 under matrix multiplication.

15. All $n \times n$ upper-triangular matrices under matrix multiplication.

16. All $n \times n$ upper-triangular matrices under matrix addition.

17. All $n \times n$ upper-triangular matrices with determinant 1 under matrix multiplication.

18. All $n \times n$ matrices with determinant either 1 or -1 under matrix multiplication.

19. Let S be the set of all real numbers except -1. Define $*$ on S by

$$a * b = a + b + ab.$$

a. Show that $*$ gives a binary operation on S.

b. Show that $\langle S, * \rangle$ is a group.

c. Find the solution of the equation $2 * x * 3 = 7$ in S.

20. This exercise shows that there are two nonisomorphic group structures on a set of 4 elements.

Let the set be $\{e, a, b, c\}$, with e the identity element for the group operation. A group table would then have to start in the manner shown in Table 4.22. The square indicated by the question mark cannot be filled in with a. It must be filled in either with the identity element e or with an element different from both e and a. In this latter case, it is no loss of generality to assume that this element is b. If this square is filled in with e, the table can then be completed in two ways to give a group. Find these two tables. (You need not check the associative law.) If this square is filled in with b, then the table can only be completed in one way to give a group. Find this table. (Again, you need not check the associative law.) Of the three tables you now have, two give isomorphic groups. Determine which two tables these are, and give the one-to-one onto renaming function which is an isomorphism.

a. Are all groups of 4 elements commutative?

b. Which table gives a group isomorphic to the group U_4, so that we know the binary operation defined by the table is associative?

c. Show that the group given by one of the other tables is structurally the same as the group in Exercise 14 for one particular value of n, so that we know that the operation defined by that table is associative also.

21. According to Exercise 12 of Section 2, there are 16 possible binary operations on a set of 2 elements. How many of these give a structure of a group? How many of the 19,683 possible binary operations on a set of 3 elements give a group structure?

Concepts

22. Consider our axioms $\mathscr{G}_1$, $\mathscr{G}_2$, and $\mathscr{G}_3$ for a group. We gave them in the order $\mathscr{G}_1\mathscr{G}_2\mathscr{G}_3$. Conceivable other orders to state the axioms are $\mathscr{G}_1\mathscr{G}_3\mathscr{G}_2$, $\mathscr{G}_2\mathscr{G}_1\mathscr{G}_3$, $\mathscr{G}_2\mathscr{G}_3\mathscr{G}_1$, $\mathscr{G}_3\mathscr{G}_1\mathscr{G}_2$, and $\mathscr{G}_3\mathscr{G}_2\mathscr{G}_1$. Of these six possible

orders, exactly three are acceptable for a definition. Which orders are not acceptable, and why? (Remember this. Most instructors ask the student to define a group on at least one test.)

4.22 Table

$*$	e	a	b	c
e	e	a	b	c
a	a	?		
b	b			
c	c			

23. The following "definitions" of a group are taken verbatim, including spelling and punctuation, from papers of students who wrote a bit too quickly and carelessly. Criticize them.

a. A group G is a set of elements together with a binary operation $*$ such that the following conditions are satisfied

$*$ is associative

There exists $e \in G$ such that

$$e * x = x * e = x = \text{identity}.$$

For every $a \in G$ there exists an a' (inverse) such that

$$a \cdot a' = a' \cdot a = e$$

b. A group is a set G such that

The operation on G is associative.

there is an identity element (e) in G.

for every $a \in G$, there is an a' (inverse for each element)

c. A group is a set with a binary operation such

the binary operation is defined

an inverse exists

an identity element exists

d. A set G is called a group over the binery operation $*$ such that for all $a, b \in G$

Binary operation $*$ is associative under addition

there exist an element $\{e\}$ such that

$$a * e = e * a = e$$

Fore every element a there exists an element a' such that

$$a * a' = a' * a = e$$

24. Give a table for a binary operation on the set $\{e, a, b\}$ of three elements satisfying axioms $\mathscr{G}_2$ and $\mathscr{G}_3$ for a group but not axiom $\mathscr{G}_1$.

25. Mark each of the following true or false.

_____ **a.** A group may have more than one identity element.

_____ **b.** Any two groups of three elements are isomorphic.

_____ **c.** In a group, each linear equation has a solution.

_____ **d.** The proper attitude toward a definition is to memorize it so that you can reproduce it word for word as in the text.

_____ **e.** Any definition a person gives for a group is correct provided that everything that is a group by that person's definition is also a group by the definition in the text.

_____ ***f.*** *Any definition a person gives for a group is correct provided he or she can show that everything* that satisfies the definition satisfies the one in the text and conversely.

_____ **g.** Every finite group of at most three elements is abelian.

_____ **h.** An equation of the form $a * x * b = c$ always has a unique solution in a group.

_____ **i.** The empty set can be considered a group.

_____ **j.** Every group is a binary algebraic structure.

Proof synopsis

We give an example of a proof synopsis. Here is a one-sentence synopsis of the proof that the inverse of an element a in a group $\langle G, * \rangle$ is unique.

> Assuming that $a * a' = e$ and $a * a'' = e$, apply the left cancellation law to the equation $a * a' = a * a''$.

Note that we said "the left cancellation law" and not "Theorem 4.15." We always suppose that our synopsis was given as an explanation given during a conversation at lunch, with no reference to text numbering and as little notation as is practical.

26. Give a one-sentence synopsis of the proof of the left cancellation law in Theorem 4.15.

27. Give at most a two-sentence synopsis of the proof in Theorem 4.16 that an equation $ax = b$ has a unique solution in a group.

Theory

28. From our intuitive grasp of the notion of isomorphic groups, it should be clear that if $\phi : G \to G'$ is a group isomorphism, then $\phi(e)$ is the identity e' of G'. Recall that Theorem 3.14 gave a proof of this for isomorphic binary structures $\langle S, * \rangle$ and $\langle S', *' \rangle$. Of course, this covers the case of groups.

It should also be intuitively clear that if a and a' are inverse pairs in G, then $\phi(a)$ and $\phi(a')$ are inverse pairs in G', that is, that $\phi(a)' = \phi(a')$. Give a careful proof of this for a skeptic who can't see the forest for all the trees.

29. Show that if G is a finite group with identity e and with an even number of elements, then there is $a \neq e$ in G such that $a * a = e$.

30. Let $\mathbb{R}^*$ be the set of all real numbers except 0. Define $*$ on $\mathbb{R}^*$ by letting $a * b = |a|b$.

a. Show that $*$ gives an associative binary operation on $\mathbb{R}^*$.

b. Show that there is a left identity for $*$ and a right inverse for each element in $\mathbb{R}^*$.

c. Is $\mathbb{R}^*$ with this binary operation a group?

d. Explain the significance of this exercise.

31. If $*$ is a binary operation on a set S, an element x of S is an **idempotent for** $*$ if $x * x = x$. Prove that a group has exactly one idempotent element. (You may use any theorems proved so far in the text.)

32. Show that every group G with identity e and such that $x * x = e$ for all $x \in G$ is abelian. [*Hint:* Consider $(a * b) * (a * b)$.]

33. Let G be an abelian group and let $c^n = c * c * \cdots * c$ for n factors c, where $c \in G$ and $n \in \mathbb{Z}^+$. Give a mathematical induction proof that $(a * b)^n = (a^n) * (b^n)$ for all $a, b \in G$.

34. Let G be a group with a finite number of elements. Show that for any $a \in G$, there exists an $n \in \mathbb{Z}^+$ such that $a^n = e$. See Exercise 33 for the meaning of a^n. [*Hint:* Consider $e, a, a^2, a^3, \ldots, a^m$, where m is the number of elements in G, and use the cancellation laws.]

35. Show that if $(a * b)^2 = a^2 * b^2$ for a and b in a group G, then $a * b = b * a$. See Exercise 33 for the meaning of a^2.

36. Let G be a group and let $a, b \in G$. Show that $(a * b)' = a' * b'$ if and only if $a * b = b * a$.

37. Let G be a group and suppose that $a * b * c = e$ for $a, b, c \in G$. Show that $b * c * a = e$ also.

38. Prove that a set G, together with a binary operation $*$ on G satisfying the left axioms 1, 2, and 3 given on page 43, is a group.

39. Prove that a nonempty set G, together with an associative binary operation $*$ on G such that

$$a * x = b \text{ and } y * a = b \text{ have solutions in } G \text{ for all } a, b \in G,$$

is a group. [*Hint:* Use Exercise 38.]

40. Let $\langle G, \cdot \rangle$ be a group. Consider the binary operation $*$ on the set G defined by

$$a * b = b \cdot a$$

for $a, b \in G$. Show that $\langle G, * \rangle$ is a group and that $\langle G, * \rangle$ is actually isomorphic to $\langle G, \cdot \rangle$. [*Hint:* Consider the map ϕ with $\phi(a) = a'$ for $a \in G$.]

41. Let G be a group and let g be one fixed element of G. Show that the map i_g, such that $i_g(x) = gxg'$ for $x \in G$, is an isomorphism of G with itself.

SECTION 5 SUBGROUPS

Notation and Terminology

It is time to explain some conventional notation and terminology used in group theory. Algebraists as a rule do not use a special symbol $*$ to denote a binary operation different from the usual addition and multiplication. They stick with the conventional additive or multiplicative notation and even call the operation *addition* or *multiplication,* depending on the symbol used. The symbol for addition is, of course, $+$, and usually multiplication is denoted by juxtaposition without a dot, if no confusion results. Thus in place of the notation $a * b$, we shall be using either $a + b$ to be read "the *sum* of a and b," or ab to be read "the *product* of a and b." There is a sort of unwritten agreement that the symbol $+$ should be used only to designate commutative operations. Algebraists feel very uncomfortable when they see $a + b \neq b + a$. For this reason, when developing our theory in a general situation where the operation may or may not be commutative, we shall always use multiplicative notation.

Algebraists frequently use the symbol 0 to denote an additive identity element and the symbol 1 to denote a multiplicative identity element, even though they may not be actually denoting the integers 0 and 1. Of course, if they are also talking about numbers at the same time, so that confusion would result, symbols such as e or u are used as

5.1 Table

	1	a	b
1	1	a	b
a	a	b	1
b	b	1	a

identity elements. Thus a table for a group of three elements might be one like Table 5.1 or, since such a group is commutative, the table might look like Table 5.2. In general situations we shall continue to use e to denote the identity element of a group.

It is customary to denote the inverse of an element a in a group by a^{-1} in multiplicative notation and by $-a$ in additive notation. From now on, we shall be using these notations in place of the symbol a'.

Let n be a positive integer. If a is an element of a group G, written multiplicatively, we denote the product $aaa \ldots a$ for n factors a by a^n. We let a^0 be the identity element e, and denote the product $a^{-1}a^{-1}a^{-1} \ldots a^{-1}$ for n factors by a^{-n}. It is easy to see that our usual law of exponents, $a^m a^n = a^{m+n}$ for $m, n \in \mathbb{Z}$, holds. For $m, n \in \mathbb{Z}^+$, it is clear. We illustrate another type of case by an example:

$$a^{-2}a^5 = a^{-1}a^{-1}aaaaa = a^{-1}(a^{-1}a)aaaa = a^{-1}eaaaa = a^{-1}(ea)aaa$$
$$= a^{-1}aaaa = (a^{-1}a)aaa = eaaa = (ea)aa = aaa = a^3.$$

5.2 Table

$+$	0	a	b
0	0	a	b
a	a	b	0
b	b	0	a

In additive notation, we denote $a + a + a + \cdots + a$ for n summands by na, denote $(-a) + (-a) + (-a) + \cdots + (-a)$ for n summands by $-na$, and let $0a$ be the identity element. Be careful: In the notation na, the number n is in $\mathbb{Z}$, not in G. One reason we prefer to present group theory using multiplicative notation, even if G is abelian, is the confusion caused by regarding n as being in G in this notation na. No one ever misinterprets the n when it appears in an exponent.

Let us explain one more term that is used so often it merits a special definition.

5.3 Definition If G is a group, then the **order** $|G|$ of G is the number of elements in G. (Recall from Section 0 that, for any set S, $|S|$ is the cardinality of S.) ■

Subsets and Subgroups

You may have noticed that we sometimes have had groups contained within larger groups. For example, the group $\mathbb{Z}$ under addition is contained within the group $\mathbb{Q}$ under addition, which in turn is contained in the group $\mathbb{R}$ under addition. When we view the group $\langle \mathbb{Z}, + \rangle$ as contained in the group $\langle \mathbb{R}, + \rangle$, it is very important to notice that the operation $+$ on integers n and m as elements of $\langle \mathbb{Z}, + \rangle$ produces the same element $n + m$ as would result if you were to think of n and m as elements in $\langle \mathbb{R}, + \rangle$. Thus we should *not* regard the group $\langle \mathbb{Q}^+, \cdot \rangle$ as contained in $\langle \mathbb{R}, + \rangle$, even though $\mathbb{Q}^+$ is contained in $\mathbb{R}$ as a set. In this instance, $2 \cdot 3 = 6$ in $\langle \mathbb{Q}^+, \cdot \rangle$, while $2 + 3 = 5$ in $\langle \mathbb{R}, + \rangle$. We are requiring not only that the set of one group be a subset of the set of the other, but also that the group operation on the subset be the *induced operation* that assigns the same element to each ordered pair from this subset as is assigned by the group operation on the whole set.

5.4 Definition If a subset H of a group G is closed under the binary operation of G and if H with the induced operation from G is itself a group, then H is a **subgroup of** G. We shall let $H \leq G$ or $G \geq H$ denote that H is a subgroup of G, and $H < G$ or $G > H$ shall mean $H \leq G$ but $H \neq G$. ■

Thus $\langle \mathbb{Z}, + \rangle < \langle \mathbb{R}, + \rangle$ but $\langle \mathbb{Q}^+, \cdot \rangle$ is *not* a subgroup of $\langle \mathbb{R}, + \rangle$, even though as sets, $\mathbb{Q}^+ \subset \mathbb{R}$. Every group G has as subgroups G itself and $\{e\}$, where e is the identity element of G.

5.5 Definition If G is a group, then the subgroup consisting of G itself is the **improper subgroup** of G. All other subgroups are **proper subgroups**. The subgroup $\{e\}$ is the **trivial subgroup** of G. All other subgroups are **nontrivial**. ■

We turn to some illustrations.

5.6 Example Let $\mathbb{R}^n$ be the additive group of all n-component row vectors with real number entries. The subset consisting of all of these vectors having 0 as entry in the first component is a subgroup of $\mathbb{R}^n$. ▲

5.7 Example $\mathbb{Q}^+$ under multiplication is a proper subgroup of $\mathbb{R}^+$ under multiplication. ▲

5.8 Example The nth roots of unity in $\mathbb{C}$ form a subgroup U_n of the group $\mathbb{C}^*$ of nonzero complex numbers under multiplication. ▲

5.9 Example There are two different types of group structures of order 4 (see Exercise 20 of Section 4). We describe them by their group tables (Tables 5.10 and 5.11). The group V is the **Klein 4-group,** and the notation V comes from the German word *Vier* for four. The group $\mathbb{Z}_4$ is isomorphic to the group $U_4 = \{1, i, -1, -i\}$ of fourth roots of unity under multiplication.

The only nontrivial proper subgroup of $\mathbb{Z}_4$ is $\{0, 2\}$. Note that $\{0, 3\}$ is *not* a subgroup of $\mathbb{Z}_4$, since $\{0, 3\}$ is *not closed* under $+$. For example, $3 + 3 = 2$, and $2 \notin \{0, 3\}$. However, the group V has three nontrivial proper subgroups, $\{e, a\}$, $\{e, b\}$, and $\{e, c\}$. Here $\{e, a, b\}$ is *not* a subgroup, since $\{e, a, b\}$ is not closed under the operation of V because $ab = c$, and $c \notin \{e, a, b\}$. ▲

5.10 Table

$\mathbb{Z}_4$:

$+$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

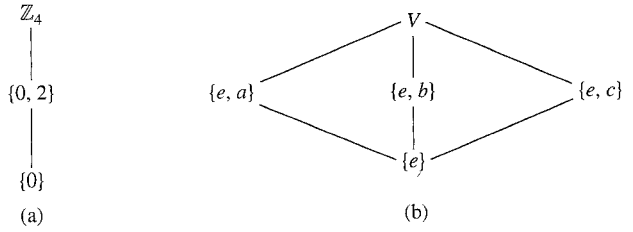
5.11 Table

V:

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

It is often useful to draw a *subgroup diagram* of the subgroups of a group. In such a diagram, a line running downward from a group G to a group H means that H is a subgroup of G. Thus the larger group is placed nearer the top of the diagram. Figure 5.12 contains the subgroup diagrams for the groups $\mathbb{Z}_4$ and V of Example 5.9.

Note that if $H \leq G$ and $a \in H$, then by Theorem 4.16, the equation $ax = a$ must have a unique solution, namely the identity element of H. But this equation can also be viewed as one in G, and we see that this unique solution must also be the identity element e of G. A similar argument then applied to the equation $ax = e$, viewed in both H and G, shows that the inverse a^{-1} of a in G is also the inverse of a in the subgroup H.



5.12 Figure (a) Subgroup diagram for $\mathbb{Z}_4$. (b) Subgroup diagram for V.

5.13 Example Let F be the group of all real-valued functions with domain $\mathbb{R}$ under addition. The subset of F consisting of those functions that are continuous is a subgroup of F, for the sum of continuous functions is continuous, the function f where $f(x) = 0$ for all x is continuous and is the additive identity element, and if f is continuous, then $-f$ is continuous. ▲

It is convenient to have routine steps for determining whether a subset of a group G is a subgroup of G. Example 5.13 indicates such a routine, and in the next theorem, we demonstrate carefully its validity. While more compact criteria are available, involving only one condition, we prefer this more transparent theorem for a first course.

5.14 Theorem A subset H of a group G is a subgroup of G if and only if

1. H is closed under the binary operation of G,
2. the identity element e of G is in H,
3. for all $a \in H$ it is true that $a^{-1} \in H$ also.

Proof The fact that if $H \leq G$ then Conditions 1, 2, and 3 must hold follows at once from the definition of a subgroup and from the remarks preceding Example 5.13.

Conversely, suppose H is a subset of a group G such that Conditions 1, 2, and 3 hold. By 2 we have at once that $\mathscr{G}_2$ is satisfied. Also $\mathscr{G}_3$ is satisfied by 3. It remains to check the associative axiom, $\mathscr{G}_1$. But surely for all $a, b, c \in H$ it is true that $(ab)c = a(bc)$ in H, for we may actually view this as an equation in G, where the associative law holds. Hence $H \leq G$. ◆

5.15 Example Let F be as in Example 5.13. The subset of F consisting of those functions that are differentiable is a subgroup of F, for the sum of differentiable functions is differentiable, the constant function 0 is differentiable, and if f is differentiable, then $-f$ is differentiable. ▲

5.16 Example Recall from linear algebra that every square matrix A has associated with it a number $\det(A)$ called its determinant, and that A is invertible if and only if $\det(A) \neq 0$. If A and B are square matrices of the same size, then it can be shown that $\det(AB) = \det(A) \cdot \det(B)$. Let G be the multiplicative group of all invertible $n \times n$ matrices with entries in $\mathbb{C}$ and let T be the subset of G consisting of those matrices with determinant 1. The equation $\det(AB) = \det(A) \cdot \det(B)$ shows that T is closed under matrix multiplication. Recall that the identity matrix I_n has determinant 1. From the equation $\det(A) \cdot \det(A^{-1}) = \det(AA^{-1}) = \det(I_n) = 1$, we see that if $\det(A) = 1$, then $\det(A^{-1}) = 1$. Theorem 5.14 then shows that T is a subgroup of G. ▲

Cyclic Subgroups

Let us see how large a subgroup H of $\mathbb{Z}_{12}$ would have to be if it contains 3. It would have to contain the identity element 0 and $3 + 3$, which is 6. Then it has to contain $6 + 3$, which is 9. Note that the inverse of 3 is 9 and the inverse of 6 is 6. It is easily checked that $H = \{0, 3, 6, 9\}$ is a subgroup of $\mathbb{Z}_{12}$, and it is the smallest subgroup containing 3.

Let us imitate this reasoning in a general situation. As we remarked before, for a general argument we always use multiplicative notation. Let G be a group and let $a \in G$. A subgroup of G containing a must, by Theorem 5.14, contain a^n, the result of computing products of a and itself for n factors for every positive integer n. These positive integral powers of a do give a set closed under multiplication. It is possible, however, that the inverse of a is not in this set. Of course, a subgroup containing a must also contain a^{-1}, and, in general, it must contain a^{-m} for all $m \in \mathbb{Z}^+$. It must contain the identity element $e = a^0$. Summarizing, *a subgroup of G containing the element a must contain all elements a^n (or na for additive groups) for all $n \in \mathbb{Z}$.* That is, a subgroup containing a must contain $\{a^n | n \in \mathbb{Z}\}$. Observe that these powers a^n of a need not be distinct. For example, in the group V of Example 5.9,

$$a^2 = e, \quad a^3 = a, \quad a^4 = e, \quad a^{-1} = a, \qquad \text{and so on.}$$

We have almost proved the next theorem.

5.17 Theorem Let G be a group and let $a \in G$. Then

$$H = \{a^n \mid n \in \mathbb{Z}\}$$

is a subgroup of G and is the smallest[†] subgroup of G that contains a, that is, every subgroup containing a contains H.

[†] We may find occasion to distinguish between the terms *minimal* and *smallest* as applied to subsets of a set S that have some property. A subset H of S is minimal with respect to the property if H has the property, and no subset $K \subset H$, $K \neq H$, has the property. If H has the property and $H \subseteq K$ for every subset K with the property, then H is the smallest subset with the property. There may be many minimal subsets, but there can be only one smallest subset. To illustrate, $\{e, a\}$, $\{e, b\}$, and $\{e, c\}$ are all minimal nontrivial subgroups of the group V. (See Fig. 5.12.) However, V contains no smallest nontrivial subgroup.

Proof We check the three conditions given in Theorem 5.14 for a subset of a group to give a subgroup. Since $a^r a^s = a^{r+s}$ for $r, s \in \mathbb{Z}$, we see that the product in G of two elements of H is again in H. Thus H is closed under the group operation of G. Also $a^0 = e$, so $e \in H$, and for $a^r \in H$, $a^{-r} \in H$ and $a^{-r}a^r = e$. Hence all the conditions are satisfied, and $H \leq G$.

Our arguments prior to the statement of the theorem showed that any subgroup of G containing a must contain H, so H is the smallest subgroup of G containing a. ◆

5.18 Definition Let G be a group and let $a \in G$. Then the subgroup $\{a^n \mid n \in \mathbb{Z}\}$ of G, characterized in Theorem 5.17, is called the **cyclic subgroup of G generated by a**, and denoted by $\langle a \rangle$. ■

5.19 Definition An element a of a group G **generates** G and is a **generator for** G if $\langle a \rangle = G$. A group G is **cyclic** if there is some element a in G that generates G. ■

5.20 Example Let $\mathbb{Z}_4$ and V be the groups of Example 5.9. Then $\mathbb{Z}_4$ is cyclic and both 1 and 3 are generators, that is,

$$\langle 1 \rangle = \langle 3 \rangle = \mathbb{Z}_4.$$

However, V is *not* cyclic, for $\langle a \rangle$, $\langle b \rangle$, and $\langle c \rangle$ are proper subgroups of two elements. Of course, $\langle e \rangle$ is the trivial subgroup of one element. ▲

5.21 Example The group $\mathbb{Z}$ under addition is a cyclic group. Both 1 and -1 are generators for this group, and they are the only generators. Also, for $n \in \mathbb{Z}^+$, the group $\mathbb{Z}_n$ under addition modulo n is cyclic. If $n > 1$, then both 1 and $n - 1$ are generators, but there may be others. ▲

5.22 Example Consider the group $\mathbb{Z}$ under addition. Let us find $\langle 3 \rangle$. Here the notation is additive, and $\langle 3 \rangle$ must contain

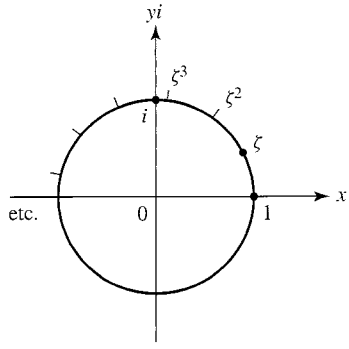
$$\begin{array}{llll} & 3, \quad 3+3=6, & 3+3+3=9, & \text{and so on,} \\ 0, \quad -3, & -3+-3=-6, & -3+-3+-3=-9, & \text{and so on.} \end{array}$$

In other words, the cyclic subgroup generated by 3 consists of all multiples of 3, positive, negative, and zero. We denote this subgroup by $3\mathbb{Z}$ as well as $\langle 3 \rangle$. In a similar way, we shall let $n\mathbb{Z}$ be the cyclic subgroup $\langle n \rangle$ of $\mathbb{Z}$. Note that $6\mathbb{Z} < 3\mathbb{Z}$. ▲

5.23 Example For each positive integer n, let U_n be the multiplicative group of the nth roots of unity in $\mathbb{C}$. These elements of U_n can be represented geometrically by equally spaced points on a circle about the origin, as illustrated in Fig. 5.24. The heavy point represents the number

$$\zeta = \cos\frac{2\pi}{n} + i \sin\frac{2\pi}{n}.$$

The geometric interpretation of multiplication of complex numbers, explained in Section 1, shows at once that as ζ is raised to powers, it works its way counterclockwise around the circle, landing on each of the elements of U_n in turn. Thus U_n under multiplication is a cyclic group, and ζ is a generator. The group U_n is the cyclic subgroup $\langle \zeta \rangle$ of the group U of all complex numbers z, where $|z| = 1$, under multiplication. ▲



5.24 Figure

EXERCISES 5

Computations

In Exercises 1 through 6, determine whether the given subset of the complex numbers is a subgroup of the group $\mathbb{C}$ of complex numbers under addition.

1. $\mathbb{R}$ **2.** $\mathbb{Q}^+$ **3.** $7\mathbb{Z}$

4. The set $i\mathbb{R}$ of pure imaginary numbers including 0

5. The set $\pi\mathbb{Q}$ of rational multiples of π **6.** The set $\{\pi^n \mid n \in \mathbb{Z}\}$

7. Which of the sets in Exercises 1 through 6 are subgroups of the group $\mathbb{C}^*$ of nonzero complex numbers under multiplication?

In Exercises 8 through 13, determine whether the given set of invertible $n \times n$ matrices with real number entries is a subgroup of $GL(n, \mathbb{R})$.

8. The $n \times n$ matrices with determinant 2

9. The diagonal $n \times n$ matrices with no zeros on the diagonal

10. The upper-triangular $n \times n$ matrices with no zeros on the diagonal

11. The $n \times n$ matrices with determinant -1

12. The $n \times n$ matrices with determinant -1 or 1

13. The set of all $n \times n$ matrices A such that $(A^T)A = I_n$. [These matrices are called **orthogonal.** Recall that A^T, the *transpose* of A, is the matrix whose jth column is the jth row of A for $1 \leq j \leq n$, and that the transpose operation has the property $(AB)^T = (B^T)(A^T)$.]

Let F be the set of all real-valued functions with domain $\mathbb{R}$ and let $\tilde{F}$ be the subset of F consisting of those functions that have a nonzero value at every point in $\mathbb{R}$. In Exercises 14 through 19, determine whether the given subset of F with the induced operation is (a) a subgroup of the group F under addition, (b) a subgroup of the group $\tilde{F}$ under multiplication.

14. The subset $\tilde{F}$

15. The subset of all $f \in F$ such that $f(1) = 0$

16. The subset of all $f \in \tilde{F}$ such that $f(1) = 1$

17. The subset of all $f \in \tilde{F}$ such that $f(0) = 1$

18. The subset of all $f \in \tilde{F}$ such that $f(0) = -1$

19. The subset of all constant functions in F.

20. Nine groups are given below. Give a *complete* list of all subgroup relations, of the form $G_i \leq G_j$, that exist between these given groups $G_1, G_2, \cdots, G_9$.
$G_1 = \mathbb{Z}$ under addition
$G_2 = 12\mathbb{Z}$ under addition
$G_3 = \mathbb{Q}^+$ under multiplication
$G_4 = \mathbb{R}$ under addition
$G_5 = \mathbb{R}^+$ under multiplication
$G_6 = \{\pi^n \mid n \in \mathbb{Z}\}$ under multiplication
$G_7 = 3\mathbb{Z}$ under addition
$G_8 =$ the set of all integral multiples of 6 under addition
$G_9 = \{6^n \mid n \in \mathbb{Z}\}$ under multiplication

21. Write at least 5 elements of each of the following cyclic groups.

a. $25\mathbb{Z}$ under addition

b. $\{(\frac{1}{2})^n \mid n \in \mathbb{Z}\}$ under multiplication

c. $\{\pi^n \mid n \in \mathbb{Z}\}$ under multiplication

In Exercises 22 through 25, describe all the elements in the cyclic subgroup of $GL(2, \mathbb{R})$ generated by the given 2×2 matrix.

22. $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$ **23.** $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ **24.** $\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$ **25.** $\begin{bmatrix} 0 & -2 \\ -2 & 0 \end{bmatrix}$

26. Which of the following groups are cyclic? For each cyclic group, list all the generators of the group.

$$G_1 = \langle \mathbb{Z}, + \rangle \quad G_2 = \langle \mathbb{Q}, + \rangle \quad G_3 = \langle \mathbb{Q}^+, \cdot \rangle \quad G_4 = \langle 6\mathbb{Z}, + \rangle$$

$G_5 = \{6^n \mid n \in \mathbb{Z}\}$ under multiplication

$G_6 = \{a + b\sqrt{2} \mid a, b \in \mathbb{Z}\}$ under addition

In Exercises 27 through 35, find the order of the cyclic subgroup of the given group generated by the indicated element.

27. The subgroup of $\mathbb{Z}_4$ generated by 3

28. The subgroup of V generated by c (see Table 5.11)

29. The subgroup of U_6 generated by $\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$

30. The subgroup of U_5 generated by $\cos \frac{4\pi}{5} + i \sin \frac{4\pi}{5}$

31. The subgroup of U_8 generated by $\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}$

32. The subgroup of U_8 generated by $\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4}$

33. The subgroup of the multiplicative group G of invertible 4×4 matrices generated by

$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

34. The subgroup of the multiplicative group G of invertible 4×4 matrices generated by

$$\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

35. The subgroup of the multiplicative group G of invertible 4×4 matrices generated by

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

36. **a.** Complete Table 5.25 to give the group $\mathbb{Z}_6$ of 6 elements.

b. Compute the subgroups $\langle 0 \rangle$, $\langle 1 \rangle$, $\langle 2 \rangle$, $\langle 3 \rangle$, $\langle 4 \rangle$, and $\langle 5 \rangle$ of the group $\mathbb{Z}_6$ given in part (a).

c. Which elements are generators for the group $\mathbb{Z}_6$ of part (a)?

d. Give the subgroup diagram for the part (b) subgroups of $\mathbb{Z}_6$. (We will see later that these are all the subgroups of $\mathbb{Z}_6$.)

5.25 Table

$\mathbb{Z}_6$:

+	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2					
3	3					
4	4					
5	5					

Concepts

In Exercises 37 and 38, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

37. A *subgroup* of a group G is a subset H of G that contains the identity element e of G and also contains the inverse of each of its elements.

38. A group G is *cyclic* if and only if there exists $a \in G$ such that $G = \{a^n \mid n \in \mathbb{Z}\}$.

39. Mark each of the following true or false.

_____ **a.** The associative law holds in every group.

_____ **b.** There may be a group in which the cancellation law fails.

_____ **c.** Every group is a subgroup of itself.
_____ **d.** Every group has exactly two improper subgroups.
_____ **e.** In every cyclic group, every element is a generator.
_____ **f.** A cyclic group has a unique generator.
_____ **g.** Every set of numbers that is a group under addition is also a group under multiplication.
_____ **h.** A subgroup may be defined as a subset of a group.
_____ **i.** $\mathbb{Z}_4$ is a cyclic group.
_____ **j.** Every subset of every group is a subgroup under the induced operation.

40. Show by means of an example that it is possible for the quadratic equation $x^2 = e$ to have more than two solutions in some group G with identity e.

Theory

In Exercises 41 and 42, let $\phi : G \to G'$ be an isomorphism of a group $\langle G, * \rangle$ with a group $\langle G', *' \rangle$. Write out a proof to convince a skeptic of the intuitively clear statement.

41. If H is a subgroup of G, then $\phi[H] = \{\phi(h) \mid h \in H\}$ is a subgroup of G'. That is, an isomorphism carries subgroups into subgroups.

42. If G is cyclic, then G' is cyclic.

43. Show that if H and K are subgroups of an abelian group G, then

$$\{hk \mid h \in H \text{ and } k \in K\}$$

is a subgroup of G.

44. Find the flaw in the following argument: "Condition 2 of Theorem 5.14 is redundant, since it can be derived from 1 and 3, for let $a \in H$. Then $a^{-1} \in H$ by 3, and by 1, $aa^{-1} = e$ is an element of H, proving 2."

45. Show that a nonempty subset H of a group G is a subgroup of G if and only if $ab^{-1} \in H$ for all $a, b \in H$. (This is one of the *more compact criteria* referred to prior to Theorem 5.14)

46. Prove that a cyclic group with only one generator can have at most 2 elements.

47. Prove that if G is an abelian group, written multiplicatively, with identity element e, then all elements x of G satisfying the equation $x^2 = e$ form a subgroup H of G.

48. Repeat Exercise 47 for the general situation of the set H of all solutions x of the equation $x^n = e$ for a fixed integer $n \geq 1$ in an abelian group G with identity e.

49. Show that if $a \in G$, where G is a finite group with identity e, then there exists $n \in \mathbb{Z}^+$ such that $a^n = e$.

50. Let a nonempty finite subset H of a group G be closed under the binary operation of G. Show that H is a subgroup of G.

51. Let G be a group and let a be one fixed element of G. Show that

$$H_a = \{x \in G \mid xa = ax\}$$

is a subgroup of G.

52. Generalizing Exercise 51, let S be any subset of a group G.

a. Show that $H_S = \{x \in G \mid xs = sx \text{ for all } s \in S\}$ is a subgroup of G.

b. In reference to part (a), the subgroup H_G is the **center of** G. Show that H_G is an abelian group.

53. Let H be a subgroup of a group G. For $a, b \in G$, let $a \sim b$ if and only if $ab^{-1} \in H$. Show that $\sim$ is an equivalence relation on G.

54. For sets H and K, we define the **intersection** $H \cap K$ by

$$H \cap K = \{x \mid x \in H \text{ and } x \in K\}.$$

Show that if $H \leq G$ and $K \leq G$, then $H \cap K \leq G$. (Remember: $\leq$ denotes "is a subgroup of," not "is a subset of.")

55. Prove that every cyclic group is abelian.

56. Let G be a group and let $G_n = \{g^n \mid g \in G\}$. Under what hypothesis about G can we show that G_n is a subgroup of G?

57. Show that a group with no proper nontrivial subgroups is cyclic.

SECTION 6 Cyclic Groups

Recall the following facts and notations from Section 5. If G is a group and $a \in G$, then

$$H = \{a^n \mid n \in \mathbb{Z}\}$$

is a subgroup of G (Theorem 5.17). This group is the **cyclic subgroup** $\langle a \rangle$ **of** G **generated by** a. Also, given a group G and an element a in G, if

$$G = \{a^n \mid n \in \mathbb{Z}\},$$

then a is a **generator of** G and the group $G = \langle a \rangle$ is **cyclic.** We introduce one new bit of terminology. Let a be an element of a group G. If the cyclic subgroup $\langle a \rangle$ of G is finite, then the **order of** a is the order $|\langle a \rangle|$ of this cyclic subgroup. Otherwise, we say that a is of **infinite order.** We will see in this section that if $a \in G$ is of finite order m, then m is the smallest positive integer such that $a^m = e$.

The first goal of this section is to describe all cyclic groups and all subgroups of cyclic groups. This is not an idle exercise. We will see later that cyclic groups serve as building blocks for all sufficiently small abelian groups, in particular, for all finite abelian groups. Cyclic groups are fundamental to the understanding of groups.

Elementary Properties of Cyclic Groups

We start with a demonstration that cyclic groups are abelian.

6.1 Theorem Every cyclic group is abelian.

Proof Let G be a cyclic group and let a be a generator of G so that

$$G = \langle a \rangle = \{a^n \mid n \in \mathbb{Z}\}.$$

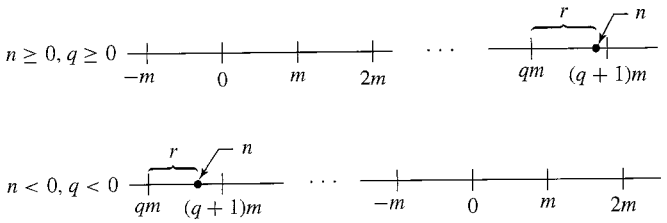
If g_1 and g_2 are any two elements of G, there exist integers r and s such that $g_1 = a^r$ and $g_2 = a^s$. Then

$$g_1 g_2 = a^r a^s = a^{r+s} = a^{s+r} = a^s a^r = g_2 g_1,$$

so G is abelian. ◆

We shall continue to use multiplicative notation for our general work on cyclic groups, even though they are abelian.

The *division algorithm* that follows is a seemingly trivial, but very fundamental tool for the study of cyclic groups.



6.2 Figure

6.3 Division Algorithm for $\mathbb{Z}$ If m is a positive integer and n is any integer, then there exist unique integers q and r such that

$$n = mq + r \qquad \text{and} \qquad 0 \leq r < m.$$

Proof We give an intuitive diagrammatic explanation, using Fig. 6.2. On the real x-axis of analytic geometry, mark off the multiples of m and the position of n. Now n falls either on a multiple qm of m and r can be taken as 0, or n falls between two multiples of m. If the latter is the case, let qm be the first multiple of m to the left of n. Then r is as shown in Fig. 6.2. Note that $0 \leq r < m$. Uniqueness of q and r follows since if n is not a multiple of m so that we can take $r = 0$, then there is a unique multiple qm of m to the left of n and at distance less than m from n, as illustrated in Fig. 6.2. ◆

In the notation of the division algorithm, we regard q as the **quotient** and r as the nonnegative **remainder** when n is divided by m.

6.4 Example Find the quotient q and remainder r when 38 is divided by 7 according to the division algorithm.

Solution The positive multiples of 7 are 7, 14, 21, 28, 35, 42, $\cdots$. Choosing the multiple to leave a nonnegative remainder less than 7, we write

$$38 = 35 + 3 = 7(5) + 3$$

so the quotient is $q = 5$ and the remainder is $r = 3$. ▲

6.5 Example Find the quotient q and remainder r when -38 is divided by 7 according to the division algorithm.

Solution The negative multiples of 7 are $-7, -14, -21, -28, -35, -42, \cdots$. Choosing the multiple to leave a nonnegative remainder less than 7, we write

$$-38 = -42 + 4 = 7(-6) + 4$$

so the quotient is $q = -6$ and the remainder is $r = 4$. ▲

We will use the division algorithm to show that a subgroup H of a cyclic group G is also cyclic. Think for a moment what we will have to do to prove this. We will have to

use the *definition* of a cyclic group since we have proved little about cyclic groups yet. That is, we will have to use the fact that G has a generating element a. We must then exhibit, in terms of this generator a, some generator $c = a^m$ for H in order to show that H is cyclic. There is really only one natural choice for the power m of a to try. Can you guess what it is before you read the proof of the theorem?

6.6 Theorem A subgroup of a cyclic group is cyclic.

Proof Let G be a cyclic group generated by a and let H be a subgroup of G. If $H = \{e\}$, then $H = \langle e \rangle$ is cyclic. If $H \neq \{e\}$, then $a^n \in H$ for some $n \in \mathbb{Z}^+$. Let m be the smallest integer in $\mathbb{Z}^+$ such that $a^m \in H$.

We claim that $c = a^m$ generates H; that is,

$$H = \langle a^m \rangle = \langle c \rangle.$$

We must show that every $b \in H$ is a power of c. Since $b \in H$ and $H \leq G$, we have $b = a^n$ for some n. Find q and r such that

$$n = mq + r \qquad \text{for} \qquad 0 \leq r < m$$

in accord with the division algorithm. Then

$$a^n = a^{mq+r} = (a^m)^q a^r,$$

so

$$a^r = (a^m)^{-q} a^n.$$

Now since $a^n \in H$, $a^m \in H$, and H is a group, both $(a^m)^{-q}$ and a^n are in H. Thus

$$(a^m)^{-q} a^n \in H; \qquad \text{that is,} \qquad a^r \in H.$$

Since m was the smallest positive integer such that $a^m \in H$ and $0 \leq r < m$, we must have $r = 0$. Thus $n = qm$ and

$$b = a^n = (a^m)^q = c^q,$$

so b is a power of c. ◆

As noted in Examples 5.21 and 5.22, $\mathbb{Z}$ under addition is cyclic and for a positive integer n, the set $n\mathbb{Z}$ of all multiples of n is a subgroup of $\mathbb{Z}$ under addition, the cyclic subgroup generated by n. Theorem 6.6 shows that these cyclic subgroups are the only subgroups of $\mathbb{Z}$ under addition. We state this as a corollary.

6.7 Corollary The subgroups of $\mathbb{Z}$ under addition are precisely the groups $n\mathbb{Z}$ under addition for $n \in \mathbb{Z}$.

This corollary gives us an elegant way to define the *greatest common divisor* of two positive integers r and s. Exercise 45 shows that $H = \{nr + ms \mid n, m \in \mathbb{Z}\}$ is a subgroup of the group $\mathbb{Z}$ under addition. Thus H must be cyclic and have a generator d, which we may choose to be positive.

6.8 Definition Let r and s be two positive integers. The positive generator d of the cyclic group

$$H = \{nr + ms \mid n, m \in \mathbb{Z}\}$$

under addition is the **greatest common divisor** (abbreviated gcd) of r and s. We write $d = \gcd(r, s)$. ■

Note from the definition that d is a divisor of both r and s since both $r = 1r + 0s$ and $s = 0r + 1s$ are in H. Since $d \in H$, we can write

$$d = nr + ms$$

for some integers n and m. We see that every integer dividing both r and s divides the right-hand side of the equation, and hence must be a divisor of d also. Thus d must be the largest number dividing both r and s; this accounts for the name given to d in Definition 6.8.

6.9 Example Find the gcd of 42 and 72.

Solution The positive divisors of 42 are 1, 2, 3, 6, 7, 14, 21, and 42. The positive divisors of 72 are 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, and 72. The greatest common divisor is 6. Note that $6 = (3)(72) + (-5)(42)$. There is an algorithm for expressing the greatest common divisor d of r and s in the form $d = nr + ms$, but we will not need to make use of it here. ▲

Two positive integers are **relatively prime** if their gcd is 1. For example, 12 and 25 are relatively prime. Note that they have no prime factors in common. In our discussion of subgroups of cyclic groups, we will need to know the following:

If r and s are relatively prime and if r divides sm, then r must divide m.	(1)

Let's prove this. If r and s are relatively prime, then we may write

$$1 = ar + bs \qquad \text{for some} \qquad a, b \in \mathbb{Z}.$$

Multiplying by m, we obtain

$$m = arm + bsm.$$

Now r divides both arm and bsm since r divides sm. Thus r is a divisor of the right-hand side of this equation, so r must divide m.

The Structure of Cyclic Groups

We can now describe all cyclic groups, up to an isomorphism.

6.10 Theorem Let G be a cyclic group with generator a. If the order of G is infinite, then G is isomorphic to $\langle \mathbb{Z}, + \rangle$. If G has finite order n, then G is isomorphic to $\langle \mathbb{Z}_n, +_n \rangle$.

Proof **Case I** *For all positive integers m, $a^m \neq e$.* In this case we claim that no two distinct exponents h and k can give equal elements a^h and a^k of G. Suppose that $a^h = a^k$ and say $h > k$. Then

$$a^h a^{-k} = a^{h-k} = e,$$

contrary to our Case I assumption. Hence every element of G can be expressed as a^m for a unique $m \in \mathbb{Z}$. The map $\phi : G \to \mathbb{Z}$ given by $\phi(a^i) = i$ is thus well defined, one to one, and onto $\mathbb{Z}$. Also,

$$\phi(a^i a^j) = \phi(a^{i+j}) = i + j = \phi(a^i) + \phi(a^j),$$

so the homomorphism property is satisfied and ϕ is an isomorphism.

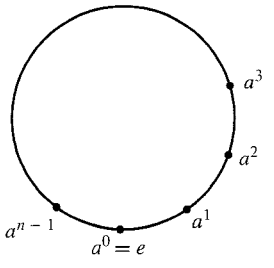
Case II *$a^m = e$ for some positive integer m.* Let n be the smallest positive integer such that $a^n = e$. If $s \in \mathbb{Z}$ and $s = nq + r$ for $0 \leq r < n$, then $a^s = a^{nq+r} = (a^n)^q a^r = e^q a^r = a^r$. As in Case 1, if $0 < k < h < n$ and $a^h = a^k$, then $a^{h-k} = e$ and $0 < h - k < n$, contradicting our choice of n. Thus the elements

$$a^0 = e, a, a^2, a^3, \cdots, a^{n-1}$$

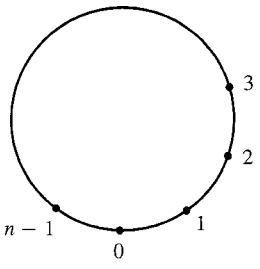
are all distinct and comprise all elements of G. The map $\psi : G \to \mathbb{Z}_n$ given by $\psi(a^i) = i$ for $i = 0, 1, 2, \cdots, n-1$ is thus well defined, one to one, and onto $\mathbb{Z}_n$. Because $a^n = e$, we see that $a^i a^j = a^k$ where $k = i +_n j$. Thus

$$\psi(a^i a^j) = i +_n j = \psi(a^i) +_n \psi(a^j),$$

so the homomorphism property is satisfied and ψ is an isomorphism. ◆



6.11 Figure



6.12 Figure

6.13 Example Motivated by our work with U_n, it is nice to visualize the elements $e = a^0, a^1, a^2, \cdots, a^{n-1}$ of a cyclic group of order n as being distributed evenly on a circle (see Fig. 6.11). The element a^h is located h of these equal units counterclockwise along the circle, measured from the bottom where $e = a^0$ is located. To multiply a^h and a^k diagrammatically, we start from a^h and go k additional units around counterclockwise. To see arithmetically

where we end up, find q and r such that

$$h + k = nq + r \qquad \text{for} \qquad 0 \leq r < n.$$

The nq takes us all the way around the circle q times, and we then wind up at a^r. ▲

Figure 6.12 is essentially the same as Fig. 6.11 but with the points labeled with the exponents on the generator. The operation on these exponents is *addition modulo n*.

Subgroups of Finite Cyclic Groups

We have completed our description of cyclic groups and turn to their subgroups. Corollary 6.7 gives us complete information about subgroups of infinite cyclic groups. Let us give the basic theorem regarding generators of subgroups for the finite cyclic groups.

6.14 Theorem Let G be a cyclic group with n elements and generated by a. Let $b \in G$ and let $b = a^s$. Then b generates a cyclic subgroup H of G containing n/d elements, where d is the greatest common divisor of n and s. Also, $\langle a^s \rangle = \langle a^t \rangle$ if and only if $\gcd(s, n) = gcd(t, n)$.

Proof That b generates a cyclic subgroup H of G is known from Theorem 5.17. We need show only that H has n/d elements. Following the argument of Case II of Theorem 6.10, we see that H has as many elements as the smallest positive power m of b that gives the identity. Now $b = a^s$, and $b^m = e$ if and only if $(a^s)^m = e$, or if and only if n divides ms. What is the smallest positive integer m such that n divides ms? Let d be the gcd of n and s. Then there exists integers u and v such that

$$d = un + vs.$$

Since d divides both n and s, we may write

$$1 = u(n/d) + v(s/d)$$

where both n/d and s/d are integers. This last equation shows that n/d and s/d are relatively prime, for any integer dividing both of them must also divide 1. We wish to find the smallest positive m such that

$$\frac{ms}{n} = \frac{m(s/d)}{(n/d)} \text{ is an integer.}$$

From the boxed division property (1), we conclude that n/d must divide m, so the smallest such m is n/d. Thus the order of H is n/d.

Taking for the moment $\mathbb{Z}_n$ as a model for a cyclic group of order n, we see that if d is a divisor of n, then the cyclic subgroup $\langle d \rangle$ of $\mathbb{Z}_n$ had n/d elements, and contains all the positive integers m less than n such that $\gcd(m, n) = d$. Thus there is only one subgroup of $\mathbb{Z}_n$ of order n/d. Taken with the preceding paragraph, this shows at once that if a is a generator of the cyclic group G, then $\langle a^s \rangle = \langle a^t \rangle$ if and only if $\gcd(s, n) = \gcd(t, n)$. ◆

6.15 Example For an example using additive notation, consider $\mathbb{Z}_{12}$, with the generator $a = 1$. Since the greatest common divisor of 3 and 12 is 3, $3 = 3 \cdot 1$ generates a subgroup of $\frac{12}{3} = 4$ elements, namely

$$\langle 3 \rangle = \{0, 3, 6, 9\}.$$

Since the gcd of 8 and 12 is 4, 8 generates a subgroup of $\frac{12}{4} = 3$ elements, namely,

$$\langle 8 \rangle = \{0, 4, 8\}.$$

Since the gcd of 12 and 5 is 1, 5 generates a subgroup of $\frac{12}{1} = 12$ elements; that is, 5 is a generator of the whole group $\mathbb{Z}_{12}$. ▲

The following corollary follows immediately from Theorem 6.14.

6.16 Corollary If a is a generator of a finite cyclic group G of order n, then the other generators of G are the elements of the form a^r, where r is relatively prime to n.

6.17 Example Let us find all subgroups of $\mathbb{Z}_{18}$ and give their subgroup diagram. All subgroups are cyclic. By Corollary 6.16, the elements 1, 5, 7, 11, 13, and 17 are all generators of $\mathbb{Z}_{18}$. Starting with 2,

$$\langle 2 \rangle = \{0, 2, 4, 6, 8, 10, 12, 14, 16\}.$$

is of order 9 and has as generators elements of the form $h2$, where h is relatively prime to 9, namely, $h = 1, 2, 4, 5, 7$, and 8, so $h2 = 2, 4, 8, 10, 14$, and 16. The element 6 of $\langle 2 \rangle$ generates $\{0, 6, 12\}$, and 12 also is a generator of this subgroup.

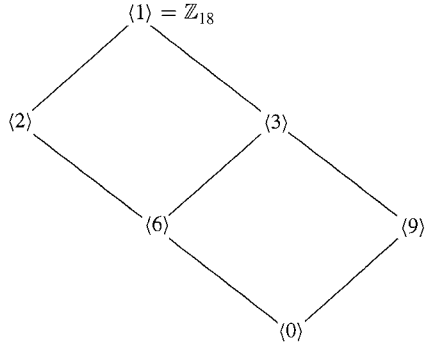
We have thus far found all subgroups generated by 0, 1, 2, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 16, and 17. This leaves just 3, 9, and 15 to consider.

$$\langle 3 \rangle = \{0, 3, 6, 9, 12, 15\},$$

and 15 also generates this group of order 6, since $15 = 5 \cdot 3$, and the gcd of 5 and 6 is 1. Finally,

$$\langle 9 \rangle = \{0, 9\}.$$

The subgroup diagram for these subgroups of $\mathbb{Z}_{18}$ is given in Fig. 6.18.



6.18 Figure Subgroup diagram for $\mathbb{Z}_{18}$.

This example is straightforward; we are afraid we wrote it out in such detail that it may look complicated. The exercises give some practice along these lines. ▲

EXERCISES 6

Computations

In Exercises 1 through 4, find the quotient and remainder, according to the division algorithm, when n is divided by m.

1. $n = 42, m = 9$

2. $n = -42, m = 9$

3. $n = -50, m = 8$

4. $n = 50, m = 8$

In Exercises 5 through 7, find the greatest common divisor of the two integers.

5. 32 and 24 **6.** 48 and 88 **7.** 360 and 420

In Exercises 8 through 11, find the number of generators of a cyclic group having the given order.

8. 5 **9.** 8 **10.** 12 **11.** 60

An isomorphism of a group with itself is an **automorphism of the group.** In Exercises 12 through 16, find the number of automorphisms of the given group.
[*Hint:* Make use of Exercise 44. What must be the image of a generator under an automorphism?]

12. $\mathbb{Z}_2$ **13.** $\mathbb{Z}_6$ **14.** $\mathbb{Z}_8$ **15.** $\mathbb{Z}$ **16.** $\mathbb{Z}_{12}$

In Exercises 17 through 21, find the number of elements in the indicated cyclic group.

17. The cyclic subgroup of $\mathbb{Z}_{30}$ generated by 25

18. The cyclic subgroup of $\mathbb{Z}_{42}$ generated by 30

19. The cyclic subgroup $\langle i \rangle$ of the group $\mathbb{C}^*$ of nonzero complex numbers under multiplication

20. The cyclic subgroup of the group $\mathbb{C}^*$ of Exercise 19 generated by $(1 + i)/\sqrt{2}$

21. The cyclic subgroup of the group $\mathbb{C}^*$ of Exercise 19 generated by $1 + i$

In Exercises 22 through 24, find all subgroups of the given group, and draw the subgroup diagram for the subgroups.

22. $\mathbb{Z}_{12}$ **23.** $\mathbb{Z}_{36}$ **24.** $\mathbb{Z}_8$

In Exercises 25 through 29, find all orders of subgroups of the given group.

25. $\mathbb{Z}_6$ **26.** $\mathbb{Z}_8$ **27.** $\mathbb{Z}_{12}$ **28.** $\mathbb{Z}_{20}$ **29.** $\mathbb{Z}_{17}$

Concepts

In Exercises 30 and 31, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

30. An element a of a group G has *order* $n \in \mathbb{Z}^+$ if and only if $a^n = e$.

31. The *greatest common divisor* of two positive integers is the largest positive integer that divides both of them.

32. Mark each of the following true or false.

_____ **a.** Every cyclic group is abelian.
_____ **b.** Every abelian group is cyclic.
_____ **c.** $\mathbb{Q}$ under addition is a cyclic group.
_____ **d.** Every element of every cyclic group generates the group.
_____ **e.** There is at least one abelian group of every finite order >0.
_____ **f.** Every group of order ≤ 4 is cyclic.

_____ **g.** All generators of $\mathbb{Z}_{20}$ are prime numbers.

_____ **h.** If G and G' are groups, then $G \cap G'$ is a group.

_____ **i.** If H and K are subgroups of a group G, then $H \cap K$ is a group.

_____ **j.** Every cyclic group of order >2 has at least two distinct generators.

In Exercises 33 through 37, either give an example of a group with the property described, or explain why no example exists.

33. A finite group that is not cyclic

34. An infinite group that is not cyclic

35. A cyclic group having only one generator

36. An infinite cyclic group having four generators

37. A finite cyclic group having four generators

The generators of the cyclic multiplicative group U_n of all nth roots of unity in $\mathbb{C}$ are the **primitive nth roots of unity**. In Exercises 38 through 41, find the primitive nth roots of unity for the given value of n.

38. $n = 4$

39. $n = 6$

40. $n = 8$

41. $n = 12$

Proof Synopsis

42. Give a one-sentence synopsis of the proof of Theorem 6.1.

43. Give at most a three-sentence synopsis of the proof of Theorem 6.6.

Theory

44. Let G be a cyclic group with generator a, and let G' be a group isomorphic to G. If $\phi : G \to G'$ is an isomorphism, show that, for every $x \in G$, $\phi(x)$ is completely determined by the value $\phi(a)$. That is, if $\phi : G \to G'$ and $\psi : G \to G'$ are two isomophisms such that $\phi(a) = \psi(a)$, then $\phi(x) = \psi(x)$ for all $x \in G$.

45. Let r and s be positive integers. Show that $\{nr + ms \mid n, m \in \mathbb{Z}\}$ is a subgroup of $\mathbb{Z}$.

46. Let a and b be elements of a group G. Show that if ab has finite order n, then ba also has order n.

47. Let r and s be positive integers.

a. Define the **least common multiple** of r and s as a generator of a certain cyclic group.

b. Under what condition is the least common multiple of r and s their product, rs?

c. Generalizing part (b), show that the product of the greatest common divisor and of the least common multiple of r and s is rs.

48. Show that a group that has only a finite number of subgroups must be a finite group.

49. Show by a counterexample that the following "converse" of Theorem 6.6 is not a theorem: "If a group G is such that every proper subgroup is cyclic, then G is cyclic."

50. Let G be a group and suppose $a \in G$ generates a cyclic subgroup of order 2 and is the *unique* such element. Show that $ax = xa$ for all $x \in G$. [*Hint:* Consider $(xax^{-1})^2$.]

51. Let p and q be distinct prime numbers. Find the number of generators of the cyclic group $\mathbb{Z}_{pq}$.

52. Let p be a prime number. Find the number of generators of the cyclic group $\mathbb{Z}_{p^r}$, where r is an integer ≥ 1.

53. Show that in a finite cyclic group G of order n, written multiplicatively, the equation $x^m = e$ has exactly m solutions x in G for each positive integer m that divides n.

54. With reference to Exercise 53, what is the situation if $1 < m < n$ and m does not divide n?

55. Show that $\mathbb{Z}_p$ has no proper nontrivial subgroups if p is a prime number.

56. Let G be an abelian group and let H and K be finite cyclic subgroups with $|H| = r$ and $|K| = s$.

a. Show that if r and s are relatively prime, then G contains a cyclic subgroup of order rs.

b. Generalizing part (a), show that G contains a cyclic subgroup of order the least common multiple of r and s.

SECTION 7 Generating Sets And Cayley Digraphs

Let G be a group, and let $a \in G$. We have described the cyclic subgroup $\langle a \rangle$ of G, which is the smallest subgroup of G that contains the element a. Suppose we want to find as small a subgroup as possible that contains both a and b for another element b in G. By Theorem 5.17, we see that any subgroup containing a and b must contain a^n and b^m for all $m, n \in \mathbb{Z}$, and consequently must contain all finite products of such powers of a and b. For example, such an expression might be $a^2b^4a^{-3}b^2a^5$. Note that we cannot "simplify" this expression by writing first all powers of a followed by the powers of b, since G may not be abelian. However, products of such expressions are again expressions of the same type. Furthermore, $e = a^0$ and the inverse of such an expression is again of the same type. For example, the inverse of $a^2b^4a^{-3}b^2a^5$ is $a^{-5}b^{-2}a^3b^{-4}a^{-2}$. By Theorem 5.14, this shows that all such products of integral powers of a and b form a subgroup of G, which surely must be the smallest subgroup containing both a and b. We call a and b **generators** of this subgroup. If this subgroup should be all of G, then we say that $\{a, b\}$ **generates** G. Of course, there is nothing sacred about taking just two elements $a, b \in G$. We could have made similar arguments for three, four, or any number of elements of G, as long as we take only finite products of their integral powers.

7.1 Example The Klein 4-group $V = \{e, a, b, c\}$ of Example 5.9 is generated by $\{a, b\}$ since $ab = c$. It is also generated by $\{a, c\}$, $\{b, c\}$, and $\{a, b, c\}$. If a group G is generated by a subset S, then every subset of G containing S generates G. ▲

7.2 Example The group $\mathbb{Z}_6$ is generated by $\{1\}$ and $\{5\}$. It is also generated by $\{2, 3\}$ since $2 + 3 = 5$, so that any subgroup containing 2 and 3 must contain 5 and must therefore be $\mathbb{Z}_6$. It is also generated by $\{3, 4\}$, $\{2, 3, 4\}$, $\{1, 3\}$, and $\{3, 5\}$, but it is not generated by $\{2, 4\}$ since $\langle 2 \rangle = \{0, 2, 4\}$ contains 2 and 4. ▲

We have given an intuitive explanation of the subgroup of a group G generated by a subset of G. What follows is a detailed exposition of the same idea approached in another way, namely via intersections of subgroups. After we get an intuitive grasp of a concept, it is nice to try to write it up as neatly as possible. We give a set-theoretic definition and generalize a theorem that was in Exercise 54 of Section 5.

7.3 Definition Let $\{S_i \mid i \in I\}$ be a collection of sets. Here I may be any set of indices. The **intersection** $\cap_{i\in I} S_i$ **of the sets** S_i is the set of all elements that are in all the sets S_i; that is,

$$\bigcap_{i\in I} S_i = \{x \mid x \in S_i \text{ for all } i \in I\}.$$

If I is finite, $I = \{1, 2, \ldots, n\}$, we may denote $\cap_{i\in I} S_i$ by

$$S_1 \cap S_2 \cap \cdots \cap S_n.$$ ■

7.4 Theorem The intersection of some subgroups H_i of a group G for $i \in I$ is again a subgroup of G.

Proof Let us show closure. Let $a \in \cap_{i\in I} H_i$ and $b \in \cap_{i\in I} H_i$, so that $a \in H_i$ for all $i \in I$ and $b \in H_i$ for all $i \in I$. Then $ab \in H_i$ for all $i \in I$, since H_i is a group. Thus $ab \in \cap_{i\in I} H_i$.

Since H_i is a subgroup for all $i \in I$, we have $e \in H_i$ for all $i \in I$, and hence $e \in \cap_{i\in I} H_i$.

Finally, for $a \in \cap_{i\in I} H_i$, we have $a \in H_i$ for all $i \in I$, so $a^{-1} \in H_i$ for all $i \in I$, which implies that $a^{-1} \in \cap_{i\in I} H_i$. ◆

Let G be a group and let $a_i \in G$ for $i \in I$. There is at least one subgroup of G containing all the elements a_i for $i \in I$, namely G is itself. Theorem 7.4 assures us that if we take the intersection of all subgroups of G containing all a_i for $i \in I$, we will obtain a subgroup H of G. This subgroup H is the smallest subgroup of G containing all the a_i for $i \in I$.

7.5 Definition Let G be a group and let $a_i \in G$ for $i \in I$. The smallest subgroup of G containing $\{a_i \mid i \in I\}$ is the **subgroup generated by** $\{a_i \mid i \in I\}$. If this subgroup is all of G, then $\{a_i \mid i \in I\}$ **generates** G and the a_i are **generators of** G. If there is a finite set $\{a_i \mid i \in I\}$ that generates G, then G is **finitely generated.** ■

Note that this definition is consistent with our previous definition of a generator for a cyclic group. Note also that the statement *a is a generator of G* may mean either that $G = \langle a \rangle$ or that a is a member of a subset of G that generates G. The context in which the statement is made should indicate which is intended. Our next theorem gives the structural insight into the subgroup of G generated by $\{a_i \mid i \in I\}$ that we discussed for two generators before Example 7.1.

7.6 Theorem If G is a group and $a_i \in G$ for $i \in I$, then the subgroup H of G generated by $\{a_i \mid i \in I\}$ has as elements precisely those elements of G that are finite products of integral powers of the a_i, where powers of a fixed a_i may occur several times in the product.

Proof Let K denote the set of all finite products of integral powers of the a_i. Then $K \subseteq H$. We need only observe that K is a subgroup and then, since H is the smallest subgroup containing a_i for $i \in I$, we will be done. Observe that a product of elements in K is again in K. Since $(a_i)^0 = e$, we have $e \in K$. For every element k in K, if we form from the product giving k a new product with the order of the a_i reversed and the opposite

sign on all exponents, we have k^{-1}, which is thus in K. For example,

$$[(a_1)^3(a_2)^2(a_1)^{-7}]^{-1} = (a_1)^7(a_2)^{-2}(a_1)^{-3},$$

which is again in K. ◆

Cayley Digraphs

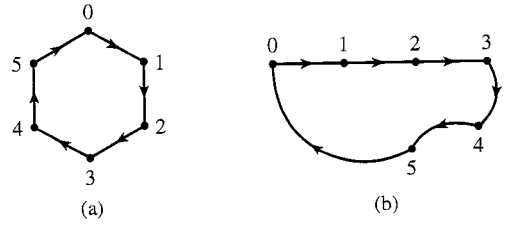
For each generating set S of a finite group G, there is a directed graph representing the group in terms of the generators in S. The term *directed graph* is usually abbreviated as *digraph*. These visual representations of groups were devised by Cayley, and are also referred to as *Cayley diagrams* in the literature.

Intuitively, a **digraph** consists of a finite number of points, called **vertices** of the digraph, and some **arcs** (each with a direction denoted by an arrowhead) joining vertices. In a digraph for a group G using a generating set S we have one vertex, represented by a dot, for each element of G. Each generator in S is denoted by one type of arc. We could use different colors for different arc types in pencil and paperwork. Since different colors are not available in our text, we use different style arcs, like solid, dashed, and dotted, to denote different generators. Thus if $S = \{a, b, c\}$ we might denote

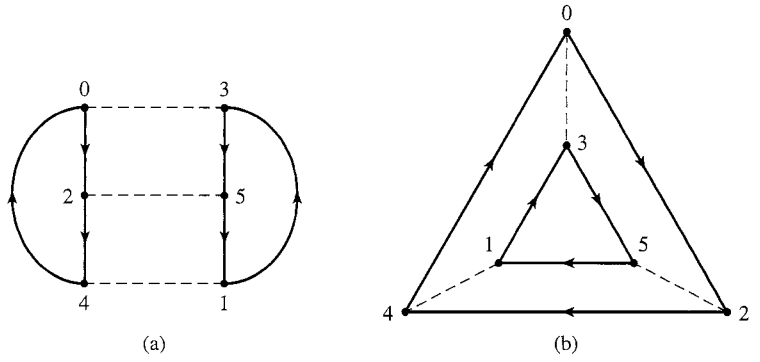
a by ——➤——, b by - - -➤- - -, and c by ·····➤·····.

With this notation, an occurrence of x•——➤——•y in a Cayley digraph means that $xa = y$. That is, traveling an arc in the direction of the arrow indicates that multiplication of the group element at the start of the arc *on the right* by the generator corresponding to that type of arc yields the group element at the end of the arc. Of course, since we are in a group, we know immediately that $ya^{-1} = x$. Thus traveling an arc in the direction opposite to the arrow corresponds to multiplication on the right by the inverse of the corresponding generator. If a generator in S is its own inverse, it is customary to denote this by omitting the arrowhead from the arc, rather than using a double arrow. For example, if $b^2 = e$, we might denote b by ————.

7.7 Example Both of the digraphs shown in Fig. 7.8 represent the group $\mathbb{Z}_6$ with generating set $S = \{1\}$. Neither the length and shape of an arc nor the angle between arcs has any significance. ▲



7.8 Figure Two digraphs for $\mathbb{Z}_6$ with $S = \{1\}$ using ——➤——$_1$.



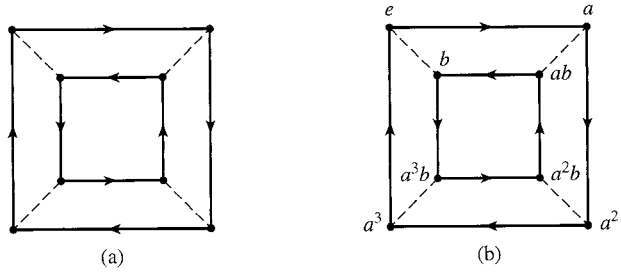
7.9 Figure Two digraphs for $\mathbb{Z}_6$ with $S = \{2, 3\}$ using $\xrightarrow[2]{}$ and $\overset{3}{\text{- - - - - -}}$.

7.10 Example Both of the digraphs shown in Fig. 7.9 represent the group $\mathbb{Z}_6$ with generating set $S = \{2, 3\}$. Since 3 is its own inverse, there is no arrowhead on the dashed arcs representing 3. Notice how different these Cayley diagrams look from those in Fig. 7.8 for the same group. The difference is due to the different choice for the set of generators. ▲

Every digraph for a group must satisfy these four properties for the reasons indicated.

Property	*Reason*
1. The digraph is connected, that is, we can get from any vertex g to any vertex h by traveling along consecutive arcs, starting at g and ending at h.	Every equation $gx = h$ has a solution in a group.
2. At most one arc goes from a vertex g to a vertex h.	The solution of $gx = h$ is unique.
3. Each vertex g has exactly one arc of each type starting at g, and one of each type ending at g.	For $g \in G$ and each generator b we can compute gb, and $(gb^{-1})b = g$.
4. If two different sequences of arc types starting from vertex g lead to the same vertex h, then those same sequences of arc types starting from any vertex u will lead to the same vertex v.	If $gq = h$ and $gr = h$, then $uq = ug^{-1}h = ur$.

It can be shown that, conversely, every digraph satisfying these four properties is a Cayley digraph for some group. Due to the symmetry of such a digraph, we can choose labels like a, b, c for the various arc types, name any vertex e to represent the identity, and name each other vertex by a product of arc labels and their inverses that we can travel to attain that vertex starting from the one that we named e. Some finite groups were first constructed (found) using digraphs.



(a) (b)

7.11 Figure

7.12 Example A digraph satisfying the four properties on page 71 is shown in Fig. 7.11 (a). To obtain Fig. 7.11 (b), we selected the labels

$\xrightarrow[a]{}$ and $\overset{}{\underset{b}{\text{- - - - - - - -}}}$,

named a vertex e, and then named the other vertices as shown. We have a group $\{e, a, a^2, a^3, b, ab, a^2b, a^3b\}$ of eight elements. Note that the vertex that we named ab could equally well be named ba^{-1}, the vertex that we named a^3 could be named a^{-1}, etc. It is not hard to compute products of elements in this group. To compute $(a^3b)(a^2b)$, we just start at the vertex labeled a^3b and then travel in succession two solid arcs and one dashed arc, arriving at the vertex a, so $(a^3b)(a^2b) = a$. In this fashion, we could write out the table for this eight-element group. ▲

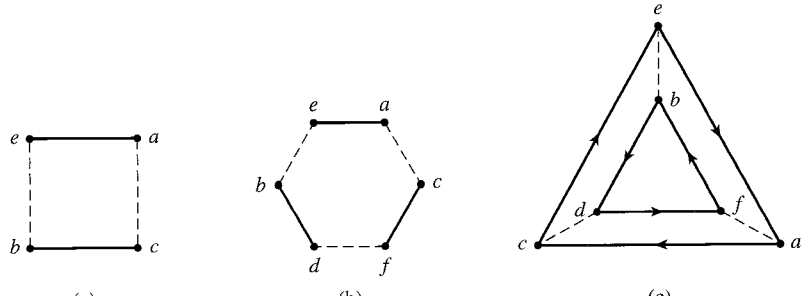
EXERCISES 7

Computations

In Exercises 1 through 6, list the elements of the subgroup generated by the given subset.

1. The subset $\{2, 3\}$ of $\mathbb{Z}_{12}$
2. The subset $\{4, 6\}$ of $\mathbb{Z}_{12}$
3. The subset $\{8, 10\}$ of $\mathbb{Z}_{18}$
4. The subset $\{12, 30\}$ of $\mathbb{Z}_{36}$
5. The subset $\{12, 42\}$ of $\mathbb{Z}$
6. The subset $\{18, 24, 39\}$ of $\mathbb{Z}$
7. For the group described in Example 7.12 compute these products, using Fig. 7.11(b).

   a. $(a^2b)a^3$ **b.** $(ab)(a^3b)$ **c.** $b(a^2b)$



(a) (b) (c)

7.13 Figure

In Exercises 8 through 10, give the table for the group having the indicated digraph. In each digraph, take e as identity element. List the identity e first in your table, and list the remaining elements alphabetically, so that your answers will be easy to check.

8. The digraph in Fig. 7.13(a)

9. The digraph in Fig. 7.13(b)

10. The digraph in Fig. 7.13(c)

Concepts

11. How can we tell from a Cayley digraph whether or not the corresponding group is commutative?

12. Referring to Exercise 11, determine whether the group corresponding to the Cayley digraph in Fig. 7.11(b) is commutative.

13. Is it obvious from a Cayley digraph of a group whether or not the group is cyclic? [*Hint:* Look at Fig. 7.9(b).]

14. The large outside triangle in Fig. 7.9(b) exhibits the cyclic subgroup $\{0, 2, 4\}$ of $\mathbb{Z}_6$. Does the smaller inside triangle similarly exhibit a cyclic subgroup of $\mathbb{Z}_6$? Why or why not?

15. The generating set $S = \{1, 2\}$ for $\mathbb{Z}_6$ contains more generators than necessary, since 1 is a generator for the group. Nevertheless, we can draw a Cayley digraph for $\mathbb{Z}_6$ with this generating set S. Draw such a Cayley digraph.

16. Draw a Cayley digraph for $\mathbb{Z}_8$ taking as generating set $S = \{2, 5\}$.

17. A **relation** on a set S of generators of a group G is an equation that equates some product of generators and their inverses to the identity e of G. For example, if $S = \{a, b\}$ and G is commutative so that $ab = ba$, then one relation is $aba^{-1}b^{-1} = e$. If, moreover, b is its own inverse, then another relation is $b^2 = e$.

a. Explain how we can find some relations on S from a Cayley digraph of G.

b. Find three relations on the set $S = \{a, b\}$ of generators for the group described by Fig. 7.11(b).

18. Draw digraphs of the two possible structurally different groups of order 4, taking as small a generating set as possible in each case. You need not label vertices.

Theory

19. Show that for $n \geq 3$, there exists a nonabelian group with $2n$ elements that is generated by two elements of order 2.